

LAND AT HESSAY, YORK

FLOOD RISK AND DRAINAGE ASSESSMENT

Final Report v1.1
March 2023

Report Title **Land at Hessay, York**
Flood Risk and Drainage Assessment
Final Report v1.1

Client Atmos Consulting

Date of issue 21 March 2023

Prepared by Keely Bonser BSc (Hons) MSc PhD MCIWEM *Director*
Tim Brook BSc (Hons) *Technical Director*

Checked and approved by Kevin Tilford BSc (Hons) MSc PhD MBA CEnv CWEM FCIWEM *Managing Director*

This document has been prepared solely as a Flood Risk and Drainage Assessment for Atmos Consulting on behalf of Hessay Solar Limited. This report is confidential to Atmos Consulting and Hessay Solar Limited and Weetwood Services Ltd accepts no responsibility or liability for any use that is made of this document other than by Atmos Consulting on behalf of Hessay Solar Limited for the purposes for which it was originally commissioned and prepared.

Contents

Signature Sheet	i
Contents	ii
List of Tables, Figures & Appendices	iii
1 Introduction.....	1
1.1 Purpose of Report	1
1.2 Structure of the Report	1
1.3 Relevant Documents and Consultation.....	1
2 Site Details and Proposed Development	2
2.1 Site Location	2
2.2 Proposed Development.....	2
2.3 Surface Waterbodies in the Vicinity of the Site.....	2
2.4 Ground Levels.....	3
2.5 Ground Conditions	4
3 Planning Policy and Guidance	5
3.1 National Planning Policy and Policy Guidance	5
3.2 Local Planning Policy	6
3.3 Drainage Technical Guidance	8
3.4 Water Framework Directive	9
3.5 Land Drainage Consent	9
4 Review of Flood Risk.....	11
4.1 Historical Records of Flooding.....	11
4.2 Flood Risk from Rivers (Fluvial)	11
4.3 Flood Risk from Small Watercourses and Surface Water (Pluvial).....	12
4.4 Flood Risk from Reservoirs, Canals and Other Water Impounding Structures	13
4.5 Flood Risk from Groundwater	13
4.6 Flood Risk Mitigation.....	13
5 Surface Water Management	15
5.1 Surface Water Drainage at the Existing Site.....	15
5.2 Surface Water Drainage at the Developed Site.....	15
6 Summary and Recommendations	19

List of Tables

Table 1:	Greenfield Runoff Rate	15
Table 2:	Summary of Proposed Surface Water Drainage Scheme.....	17
Table 3:	Maintenance Requirements	17

List of Figures

Figure 1:	Site Location.....	2
Figure 2:	Location of Surface Waterbodies.....	3
Figure 3:	Digital Terrain Model from LiDAR Data.....	3
Figure 4:	Flood Map for Planning.....	11
Figure 5:	Flood Risk from Surface Water	13
Figure 6:	Indicative Drainage Areas	16

List of Appendices

Appendix A:	Indicative Drainage Layout
Appendix B:	Ainsty (2008) Internal Drainage Board Consultation Correspondence
Appendix C:	Environment Agency Consultation Response - Flood Risk Information
Appendix D:	Greenfield Runoff Calculations
Appendix E:	Surface Water Attenuation - Storage Volume Calculations

1 INTRODUCTION

1.1 Purpose of Report

Weetwood Services Ltd (“Weetwood”) has been instructed by Atmos Consulting on behalf of Hessay Solar Limited (“the applicant”) to prepare a Flood Risk and Drainage Assessment (FRDA) report to accompany a full planning application for the proposed development of land at Hessay, York (“the site”) for use as a solar farm.

The assessment has been undertaken in accordance with the requirements of the revised National Planning Policy Framework (NPPF) updated on 20 July 2021 and the Planning Practice Guidance (PPG) updated on 25 August 2022.

1.2 Structure of the Report

The report is structured as follows:

- Section 1** Introduction and report structure
- Section 2** Provides background information relating to the development site
- Section 3** Presents national and local flood risk and drainage planning policy
- Section 4** Assesses the potential risk of flooding to the development site
- Section 5** Addresses the effect of the proposed development on surface water runoff and presents an illustrative surface water drainage scheme
- Section 6** Presents a summary of key findings and the recommendations

1.3 Relevant Documents and Consultation

The assessment has been informed by the following documents:

- Sustainable Drainage Systems Guidance for Developers, City of York Council, August 2018
- Local Plan – Publication Draft (Regulation 19 Consultation), City of York Council, February 2018
- Preliminary Flood Risk Assessment - Addendum, City of York Council, 2017
- Strategic Flood Risk Assessment, City of York Council, March 2013
- Preliminary Flood Risk Assessment, City of York Council, July 2011
- Draft Local Plan Incorporating the 4th Set of Changes – Development Control Local Plan, City of York Council, April 2005

2 SITE DETAILS AND PROPOSED DEVELOPMENT

2.1 Site Location

The approximately 61.1 ha greenfield site is located on land to the south of Mullingar Farm and Low Moor Lane, Hessay at Ordnance Survey (OS) National Grid Reference SE 532 525, as shown in **Figure 1**. The site comprises agricultural farmland.

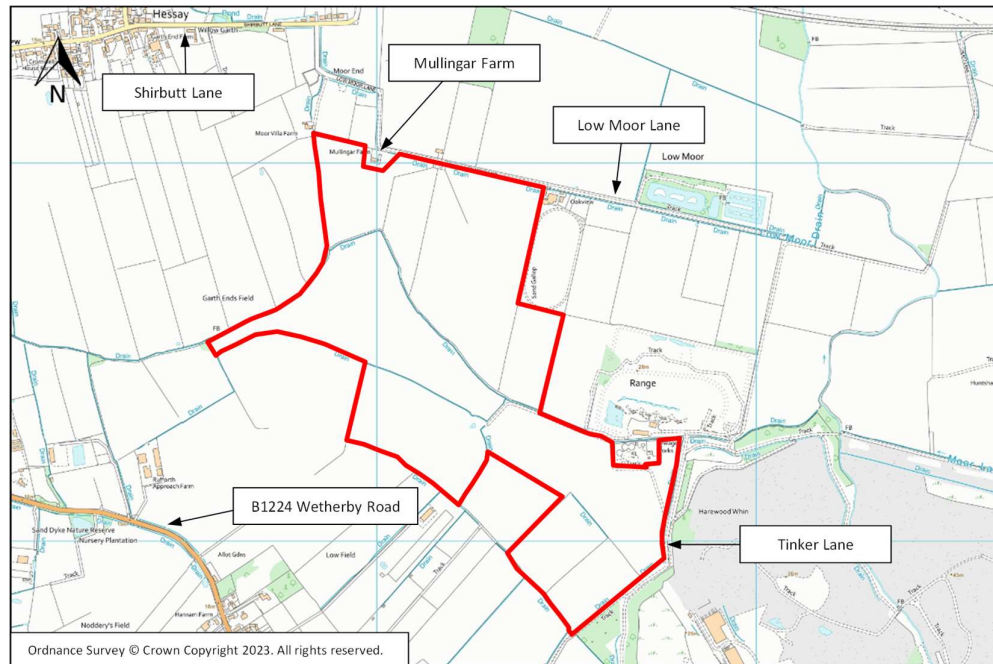


Figure 1: Site Location

2.2 Proposed Development

The proposed development comprises a solar farm comprising: 49.99 Megawatt solar array (the front and back panel heights of the panels are up to 1.0 m and 3.0 m respectively); inverter/transformer stations; substation buildings; storage containers; access tracks; security fencing; and associated landscaping across the site.

The proposed site layout plan is illustrated on the indicative drainage layout (**Appendix A**). Vehicular access will be provided via Tinker Lane and Low Moor Lane.

The proposed development would be for a lifetime period of 40 years, after which the site would be decommissioned and restored back to full agricultural use. The NPPF classifies solar farm development as Essential Infrastructure.

2.3 Surface Waterbodies in the Vicinity of the Site

There is a network of drainage ditches on and within the vicinity of the site, including Smawith Dyke, Rufforth North Dyke (B), Oak Nursery Dyke, Tinker Lane Dyke and Low Moor Lane Dyke. Most of the on-site drains fall under the jurisdiction of the Ainsty (2008) Internal Drainage Board (IDB) (refer **Figure 2**). The watercourses ultimately outfall to the River Ouse approximately 3.7 km north of the site.

Laybourne Lakes is located approximately 280 m east of the north-eastern corner of the site and a pond is located approximately 40 m to the north of the south-eastern portion of the site.

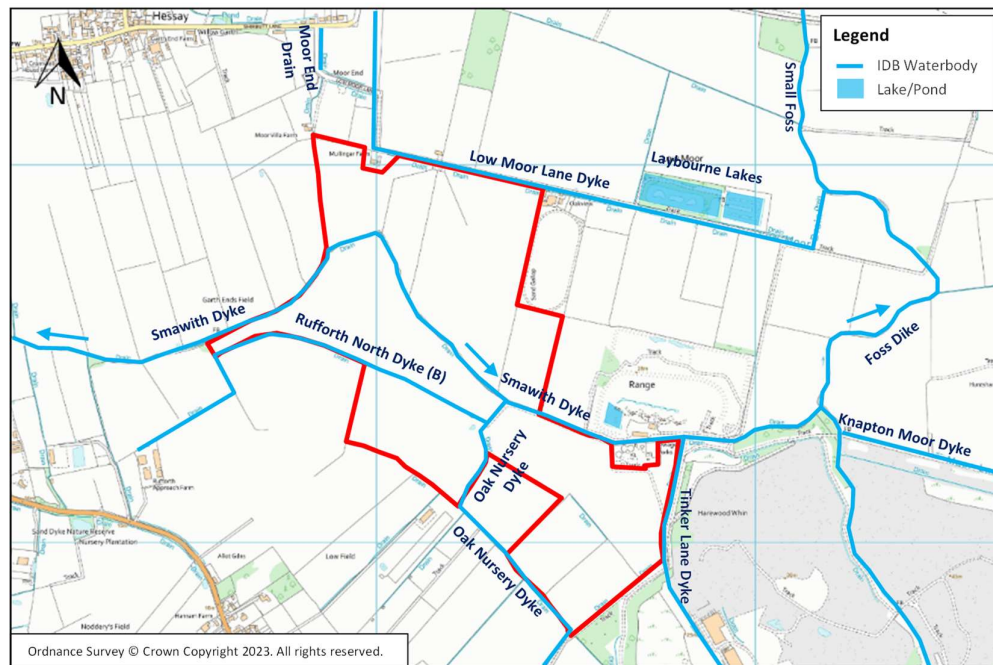


Figure 2: Location of Surface Waterbodies

2.4 Ground Levels

LiDAR data has been used to develop a digital terrain model of the site and surrounding area as illustrated in **Figure 3**. Site levels are shown to be relatively flat and in the region of 13.7 to 14.8 m AOD.

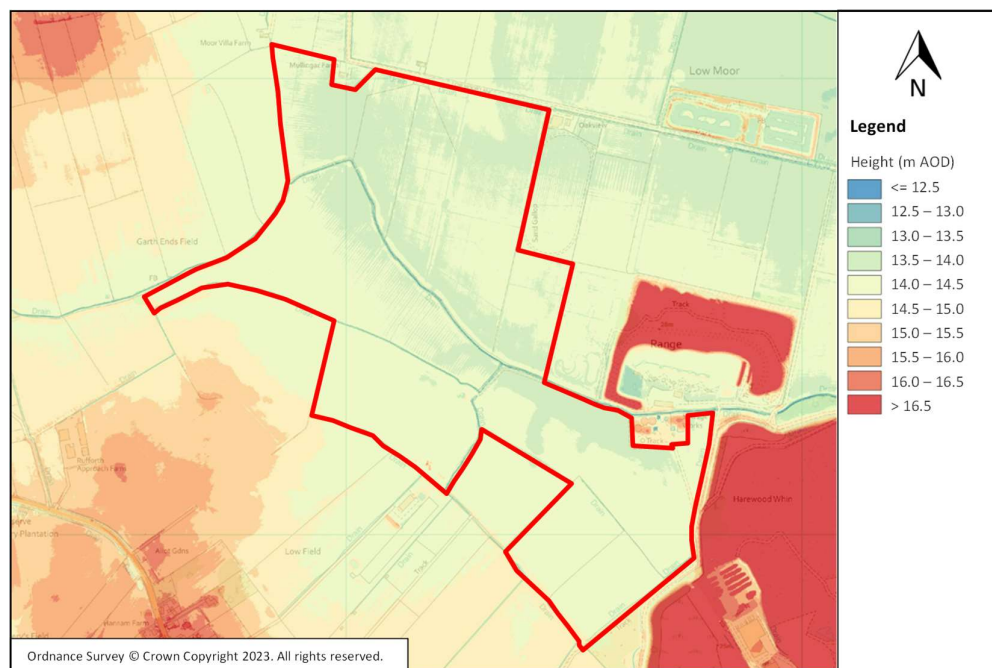


Figure 3: Digital Terrain Model from LiDAR Data

2.5 Ground Conditions

According to the Soilscales soils dataset produced by the Cranfield Soil and AgriFood Institute¹, soil conditions at the site and within the surrounding area are described as slowly permeable, seasonally wet loamy and clayey soils.

British Geological Survey mapping of surface geology² indicates the underlying bedrock formation comprises Sherwood Sandstone Group - Sandstone, overlain by superficial deposits of Alne Glaciolacustrine Formation – Clay, silty.

The National Geoscience Data Centre's Single Onshore Borehole Index² holds records of one borehole approximately 500 m to the east of the site. Ground conditions are shown to comprise of clay with bands of sand up to 17 m bgl underlain by sandstone.

According to the MAGIC website³ the superficial deposits at the site are classified as an Unproductive aquifer whilst the underlying sandstone bedrock is classified as a Principal aquifer. The site is not shown to be located within a designated groundwater source protection zone.

¹ www.landis.org.uk/soilscales/

² <https://www.bgs.ac.uk/map-viewers/geoindex-onshore/>

³ <https://magic.defra.gov.uk/MagicMap.aspx>

3 PLANNING POLICY AND GUIDANCE

3.1 National Planning Policy and Policy Guidance

The thrust of national planning policy, as articulated in the NPPF is that inappropriate development in areas at risk of flooding should be avoided where possible, as summarised below:

- Inappropriate development in areas at risk of flooding should be avoided and that development should be directed away from areas at highest risk (whether existing or future), but where development is necessary in such areas, the development should be made safe for its lifetime without increasing flood risk elsewhere (NPPF para. 159).
- The policy of seeking to steer development to areas with the lowest risk of flooding, from any source, is implemented through the application of the flood risk Sequential Test. Development should not be allocated or permitted if there are reasonably available sites, appropriate for the proposed development in areas with a lower risk of flooding. The sequential approach should be used in areas known to be at risk now or in the future from any form of flooding (NPPF para. 162).
- If it is not possible for development to be located in zones with a lower risk of flooding (taking into account wider sustainable development objectives) the Exception Test may have to be applied. The need for the test will depend on the potential vulnerability of the site and of the vulnerability of the development proposed (as set out in Annex 3 of NPPF; also PPG Table 2) (NPPF para. 163). For example, the Exception Test need not be applied for less vulnerable development in any flood zone, or for more vulnerable development in flood zones 1 or 2.
- Where the Exception Test must be applied, application of the test for development proposals at the application stage should be informed by a site-specific flood risk assessment. For the test to be passed it should be demonstrated that: (a) the development would provide wider sustainability benefits to the community that outweigh the flood risk; (b) and the development will be safe for its lifetime taking account of the vulnerability of its users, without increasing flood risk elsewhere, and, where possible, will reduce flood risk overall (NPPF para. 164). Both elements of the test should be satisfied for the development to be permitted (NPPF para. 165).
- A site-specific flood risk assessment should be provided for all development in flood zones 2 and 3 [whilst] in flood zone 1, an assessment should accompany all proposals involving: sites of 1 ha or more; land which has been identified by the Environment Agency as having critical drainage problems; land identified in a strategic flood risk assessment as being at increased flood risk in future; or land that may be subject to other sources of flooding, where its development would introduce a more vulnerable use (NPPF para. 167).
- Development should not increase flood risk elsewhere (NPPF para. 167).
- Development should only be allowed in areas at risk of flooding where the flood risk assessment (and the sequential and exception tests, as required), demonstrate that: a) within the site, the most vulnerable development is located in areas of lowest flood risk (unless there are overriding reasons to prefer a different location); b) the development is appropriately flood resistant and resilient such that, in the event of a flood, it could be quickly brought back into use without significant refurbishment; c) the development incorporates sustainable drainage systems, unless there is clear evidence that this would be inappropriate; d) any residual (flood) risk can be safely managed; and e) safe access and escape routes are included where appropriate, as part of an agreed emergency plan (NPPF para.167).
- Applications for some minor development and changes of use should not be subject to the sequential or exception tests (NPPF para. 168). The exceptions are stated in Footnote 56.
- Major development should incorporate sustainable drainage systems unless there is clear evidence that this would be inappropriate. The systems should: a) take account of advice from the lead local flood authority; b) have appropriate proposed minimum operational standards; c) have maintenance arrangements in place to ensure an acceptable standard of operation for the lifetime of the development; and d) where possible, provide multifunctional benefits (NPPF para. 169).

Guidance on application of the sequential and exception test is provided in the PPG. For example:

- The approach is designed to ensure that areas at little or no risk of flooding from any source are developed in preference to areas at higher risk. This means avoiding, so far as possible, development

in current and future (i.e. taking climate change into account) medium and high flood risk areas considering all sources of flooding including areas at risk of surface water flooding (PPG para. 023).

- Where it is not possible to locate development in low risk areas, the Sequential Test should go on to compare reasonably available sites within medium risk areas and then, only where there are no reasonably available sites in low and medium risk areas, within high risk areas (PPG para. 024).
- Initially, the presence of existing flood risk management infrastructure should be ignored, as the long-term funding, maintenance and renewal of this infrastructure is uncertain. Climate change will also impact upon the level of protection infrastructure will offer throughout the lifetime of development (PPG para. 024).
- The Sequential Test should be applied to 'Major' and 'Non-major development' proposed in areas at risk of flooding, but it will not be required where; the site has been allocated for development and subject to the test at the plan making stage (provided the proposed development is consistent with the use for which the site was allocated and provided there have been no significant changes to the known level of flood risk to the site, now or in the future which would have affected the outcome of the test); the site is in an area at low risk from all sources of flooding, unless the Strategic Flood Risk Assessment, or other information, indicates there may be a risk of flooding in the future; the application is for a development type that is exempt from the test, as specified in footnote 56 of the NPPF (PPG para. 027).
- For individual planning applications subject to the Sequential Test, the area to apply the test will be defined by local circumstances relating to the catchment area for the type of development proposed. For some developments this may be clear, for example, the catchment area for a school. In other cases, it may be identified from other Plan policies. For example, where there are large areas in Flood Zones 2 and 3 (medium to high probability of flooding) and development is needed in those areas to sustain the existing community, sites outside them are unlikely to provide reasonable alternatives. Equally, a pragmatic approach needs to be taken where proposals involve comparatively small extensions to existing premises (relative to their existing size), where it may be impractical to accommodate the additional space in an alternative location. For nationally or regionally important infrastructure the area of search to which the Sequential Test could be applied will be wider than the local planning authority boundary (PPG para. 027).
- 'Reasonably available sites' are those in a suitable location for the type of development with a reasonable prospect that the site is available to be developed at the point in time envisaged for the development. These could include a series of smaller sites and/or part of a larger site if these would be capable of accommodating the proposed development. Such lower-risk sites do not need to be owned by the applicant to be considered 'reasonably available' (PPG para. 028).
- The Exception Test should only be applied as set out in Table 2 [of the PPG ("Flood Risk Vulnerability and Flood Zone Incompatibility")] and only if the Sequential Test has shown that there are no reasonably available, lower risk sites, suitable for the proposed development, to which the development could be steered (PPG para. 032).

3.2 Local Planning Policy

The City of York Local Plan was adopted by the City of York Council in April 2005. Policy GP15a: Development and Flood Risk states:

There will be a presumption against built development (except for essential infrastructure) within the functional floodplain outside existing settlement limits.

Proposals for new built development on previously undeveloped land outside defined settlement limits will only be granted where it can be demonstrated that the development will not result in the net loss of floodplain storage capacity, not impede water flows and not increase flood risk elsewhere.

All applications in the low to medium risk² or high risk³ areas should submit a Flood Risk Assessment (FRA) providing an assessment of additional risk arising from the proposal and the measures proposed to deal with these effects. Developers must satisfy the Local Planning Authority that any flood risk will be successfully managed with the minimum environmental effect and ensure that the site can be developed, serviced and occupied safely.

The use of sustainable drainage systems to mimic natural drainage will be encouraged in all new developments in order to reduce surface water run-off.

Discharges from new development should not exceed the capacity of existing and proposed receiving sewers and watercourses and long term run-off from development sites should always be less than the level of pre development rainfall run-off.

Where required the provision and future maintenance of flood mitigation and defence measures will be sought from the developer.

¹ Low risk areas are defined (PPG25) as having an annual probability of flooding (river) less than 0.1%

² Low to medium areas of flood risk are defined (PPG25) as having an annual probability of flooding (river) 0.1-1.0%

³ High risk areas of flood risk are defined (PPG25) as having an annual probability of flooding (river) greater than 1.0%.

The City of York Local Plan – Publication Draft (February 2018) contains the following policies that are relevant in respect of flood risk and drainage:

Policy DP2: Sustainable Development states in part, that development will help conserve and enhance the environment through reducing flood risk by ensuring that new development is not subject to or does not contribute to flooding, and maintaining water quality.

Policy ENV4: Flood Risk states:

New development shall not be subject to unacceptable flood risk and shall be designed and constructed in such a way that mitigates against current and future flood events.

An assessment of whether proposed development is likely to be affected by flooding and whether it will increase flood risk locally and elsewhere in the catchment must be undertaken. The assessment of proposed development against its flood risk vulnerability and its compatibility with this vulnerability, as defined in the most up to date Strategic Flood Risk Assessment (SFRA), will determine whether development is appropriate, what detailed policies for the resultant flood zone classification, as stated in the SFRA will apply, and whether a further Exception Test (that makes provision for sites in a zone with a higher probability of flooding to be assessed against wider sustainability benefits, provided that the flood risk posed is controlled and mitigated to an acceptable level) is subsequently required.

Where flood risk is present, development will only be permitted when the local planning authority is satisfied that any flood risk within the catchment will be successfully managed (through a management and maintenance plan for the lifetime of the development) and there are details of proposed necessary mitigation measures.

A flood risk assessment must be submitted with any planning application where flood risk is an issue, regardless of its location within the flood zones. In addition, a site specific flood risk assessment that takes account of future climate change must be carried out for all planning applications of 1 hectare or greater in Flood Zone 1 and for all applications in Flood Zones 2, 3a, 3a(i) and 3b.

Areas of greater flood risk may be utilised for appropriate green infrastructure spaces.

Policy ENV5: Sustainable Drainage states:

For all development on brownfield sites, surface water flow shall be restricted to 70% of the existing runoff rate (i.e. 30% reduction in existing runoff), unless it can be demonstrated that it is not reasonably practicable to achieve this reduction in runoff.

Sufficient attenuation and long term storage should be provided to ensure surface water flow does not exceed the restricted runoff rate. Such attenuation and storage measures must accommodate at least a 1 in 30 year storm. Any design should also ensure that storm water resulting from a 1 in 100 year event plus the recommended additional flows from the latest climate change advice, to account for climate change and surcharging the drainage system, can be stored on the site without risk to people or property and without overflowing into a watercourse or adjacent areas.

Where these surface water run-off limitations are likely to be exceeded development may be approved provided sufficient facilities for the long-term storage of surface water are installed within the development or a suitable location elsewhere. Long term surface water storage facilities must not cause detriment to existing heritage and environmental assets.

For new development on greenfield sites, surface water flows arising from the development, once it is complete (and including any intermediate stages), shall be no higher than the existing rate prior to development taking place, unless it can be demonstrated that it is not reasonably practicable to achieve this.

Sustainable Drainage System (SuDS) methods of source control and water quality improvement should be utilised for all new development, to minimise the risk of pollution and to attenuate flood volumes. Such facilities should be provided on-site, or where this is not possible, close to the site.

Where new development is proposed within or adjacent to built-up areas it should be demonstrated that retrofitting existing surface water drainage systems, in those areas for flood prevention, and SuDS within the existing built environment have been explored. Any retrofitting proposals must not damage existing environmental assets including but not limited to landscapes, trees and hedgerows and agricultural land. The authority will support applications where SuDS are enhanced for Biodiversity.

In exceptional circumstances, where SuDS methods of source control and water quality cannot be provided, it must be demonstrated that:

- i. it is not possible to incorporate SuDS, either on site, or close to the site; and*
- ii. an acceptable means of surface water disposal is provided which does not increase the risk of flooding, does not damage existing environmental assets and improves on the current situation.*

Measures to restrict surface water run-off rates shall be designed and implemented to prevent an unacceptable risk to contamination of groundwater. The type of SuDS used should be appropriate to the site in question and should ensure that there is no pollution of the water environment including both ground and surface waters.

New development will not be permitted to allow ground water and/or the outflow from land drainage to enter public sewers.

Existing land drainage systems should not suffer any detriment as a result of development.

3.3 Drainage Technical Guidance

Non-statutory technical standards for sustainable drainage published by DEFRA in March 2015 set out how surface water runoff generated during the present day 1 in 30 and 1 in 100 annual exceedance probability (AEP) rainfall events and for events exceeding the present day 1 in 100 AEP event should be managed, how peak runoff rates should be restricted and how runoff volumes should be controlled.

Ainsty (2008) IDB has advised by way of an email dated 24 November 2022 (refer **Appendix B** for full response) that the Board would view any solar panels as being an “impermeable area”, that Type 1 material is also regarded as impermeable, whilst Type 3 material would be regarded as permeable.

The Board has also provided the following guidance:

Soakaways: The Board always recommends that soakaways are first considered in accordance with the PPG hierarchy for the management of surface water. The Board would therefore recommend that soakaway testing is undertaken, in accordance with BRE Digest 365, to ascertain whether the soil structure is suitable for a soakaway system. Should the testing prove to be successful the applicant should then submit a design for the soakaway, for approval by the Lead Local Flood Authority, which should:

- Accommodate a 1:30 year event with no surface flooding (plus 30% allowance for climate change); and
- Should accommodate no overland discharge off the site in a 1:100 year event (plus 30% allowance for climate change).

Discharge into a Watercourse: The Board will only accept a discharge into a watercourse (directly or indirectly) where soakaways are not feasible. The Board has advised that when there is a direct discharge to a watercourse or an indirect discharge to a watercourse it would require details regarding the following:

- a) What the applicant is proposing to discharge into (e.g. a watercourse) and the location of the proposed point of connection.
- b) If the applicant is proposing to discharge directly (or through private drainage) into a watercourse, and if that watercourse is not maintained by the Board, the Board would require (i) details of where the watercourse is flowing to (e.g. a simple plan showing the route of the watercourse to the nearest Board maintained watercourse); and (ii) details of the condition of the watercourse to ensure the same is flowing freely prior to any discharge.
- c) Where a new connection to a watercourse is proposed, the maximum discharge that will be accepted is the “greenfield” rate of 1.4 litres per second per hectare.
- d) Simple details as to what is proposed with regards to how the flow will be restricted to the agreed discharge rate.
- e) Details of the proposed surface water storage system (which we would usually recommend is impermeably lined) and the proposed storage volume and accompanying calculations, noting that the system should accommodate a 1:30 year event with no surface flooding (plus 30% allowance for climate change); and no overland discharge off the site in a 1:100 year event (plus 30% allowance for climate change). Ideally, the system should accommodate the full 1:100 year storm event (plus 30% allowance for climate change) wherever possible.
- f) Details of the proposed outfall structure into the watercourse.

3.4 Water Framework Directive

The Water Framework Directive (WFD) provides a legal framework for the protection, improvement and sustainable use of inland surface waters, groundwater, transitional waters, and coastal waters across England, and seeks to:

- Prevent deterioration in the status of aquatic ecosystems, protect them and improve the ecological condition of waters
- Achieve at least ‘good’ status for all waterbodies by 2015
- Promote the sustainable use of water as a natural resource
- Conserve habitats and species that depend directly on water
- Progressively reduce or phase out the release of individual pollutants or groups of pollutants that present a significant threat to the aquatic environment
- Progressively reduce the pollution of groundwater and prevent or limit the entry of pollutants; and
- Contribute to mitigating the effects of floods and droughts.

The WFD applies to any proposed development which has the potential to impact on a waterbody. Where this is the case, the Environment Agency may require evidence demonstrating that the proposed development does not compromise the aims of the WFD.

3.5 Land Drainage Consent

Ainsty (2008) IDB has advised by way of an email dated 24 November 2022 (refer Appendix B for full response) that under the Land Drainage Act 1991 and the Boards’ byelaws, the Board’s prior written consent (outside of the planning process) is needed for:

- Any connection into a Board maintained watercourse, or any ordinary watercourse in the Board’s district.
- Any discharge, or change in the rate of discharge, into a Board maintained watercourse, or any ordinary watercourse in the Board’s district. This applies whether the discharge enters the watercourse either directly or indirectly (i.e. via a third party asset such as a mains sewer).

- Works within or over a Board maintained watercourse, or any ordinary watercourse in the Board's district – for example, land drainage, an outfall structure, bridges, culverting etc.
- Any construction, fencing or planting within 9 metres of the top of the embankment of a Board maintained watercourse
- Undertaking activities controlled by local byelaws also requires the relevant consent.

Regarding works very close to Board maintained watercourses, the Board has advised:

- The Board's prior written consent is required for any new structures or fences or planting or hardstanding within 9 metres of the top of the embankment of any Board maintained watercourse.
- The Board would usually resist any such new structures as we must have sufficient access to these watercourses to maintain them.
- In addition, we want to ensure that there is nothing to destabilise the embankments of the watercourses.

Whilst for works very close to "ordinary watercourses" not maintained by the Board, the Board has advised:

- For any watercourses not maintained by the Board, we would be considering what works are proposed over or very close to them.
- We would usually state a minimum of 3 metres from the top of the embankment of any ordinary watercourse is left completely clear.
- Again, we would want to ensure that there is nothing that could destabilise the embankments over time and effect the flows of the watercourses.

In respect of new access tracks over watercourses. The Board has advised:

- If you are proposing to create any access tracks over a watercourse within the Board's district, the Board's prior written consent must be obtained.
- We would request full details of the proposed bridges and want to ensure that any culvert is sufficiently sized for the watercourse so there is no effect on the water flows.

Whilst, in respect of underground cables, the Board has advised:

- If you are proposing to insert a cable under a watercourse within the Board's district, the Board's prior written consent must again be obtained. In these instances, we would prefer Directional Drilling to be used.

4 REVIEW OF FLOOD RISK

4.1 Historical Records of Flooding

The Environment Agency Historic Flood Map⁴ and the 2013 Strategic Flood Risk Assessment do not hold any records of flooding at or within the immediate vicinity of the site.

However, Ainsty (2008) IDB has advised, by way of an email dated 28 November 2022 (refer **Appendix B**), that “*Smawith Dyke is known to overspill onto land adjacent and store a vast amount of water leading up to Mullingar farm house to the north. Flood water remains on the land for a short amount of time until levels/rainfall recede enough.*”

4.2 Flood Risk from Rivers (Fluvial)

The Environment Agency Flood Map for Planning (Rivers and Sea) (**Figure 4**) indicates the site to be predominantly located in flood zone 1, with the north-eastern corner of the site located in flood zone 2. The north-western boundary of the site is also indicated to abut flood zones 3 and 2. This is reiterated on Figure 10A of the 2013 Strategic Flood Risk Assessment (“All Flood Zones Plan ‘A’ - North”).

Table 1 of the PPG defines flood zones as follows:

- Flood zone 1: Low Probability. Land having a less than 1 in 1,000 annual probability of river or sea flooding
- Flood zone 2: Medium Probability. Land having between a 1 in 100 and 1 in 1,000 annual probability of river flooding or between a 1 in 200 and 1 in 1,000 annual probability of sea flooding
- Flood zone 3a: High Probability. Land having a 1 in 100 or greater annual probability of river flooding or a 1 in 200 or greater annual probability of sea flooding
- Flood zone 3b: Functional Floodplain. Land where water from rivers or the sea has to flow or be stored in times of flood. Land having a 1 in 30 or greater annual probability of flooding, with any existing flood risk management infrastructure operating effectively or land that is designed to flood (such as a flood attenuation scheme), even if it would only flood in more extreme events (such as a 1 in 1,000 annual probability of flooding).

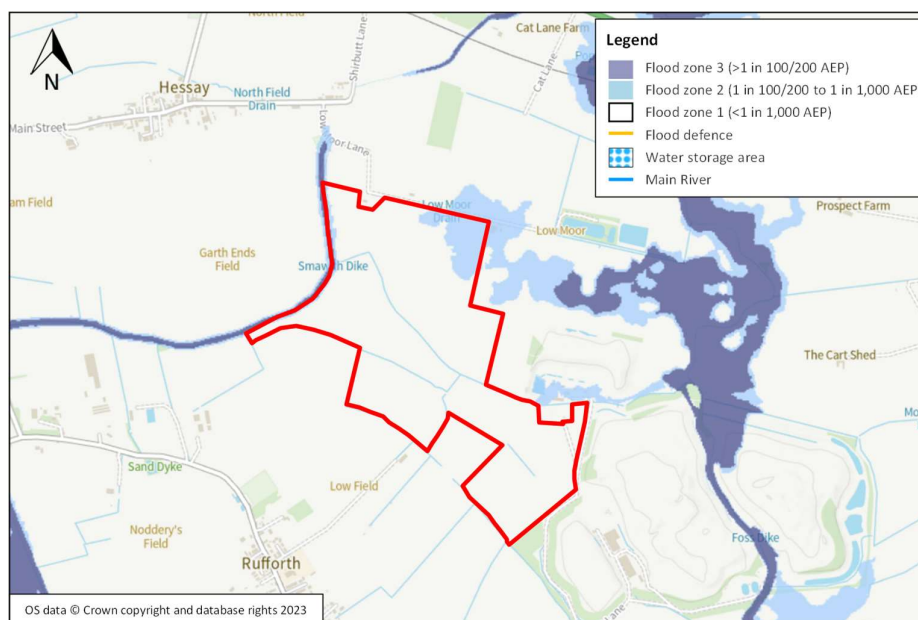


Figure 4: Flood Map for Planning
Source: gov.uk website; Accessed: February 2023

⁴ <https://data.gov.uk/dataset/76292bec-7d8b-43e8-9c98-02734fd89c81/historic-flood-map>

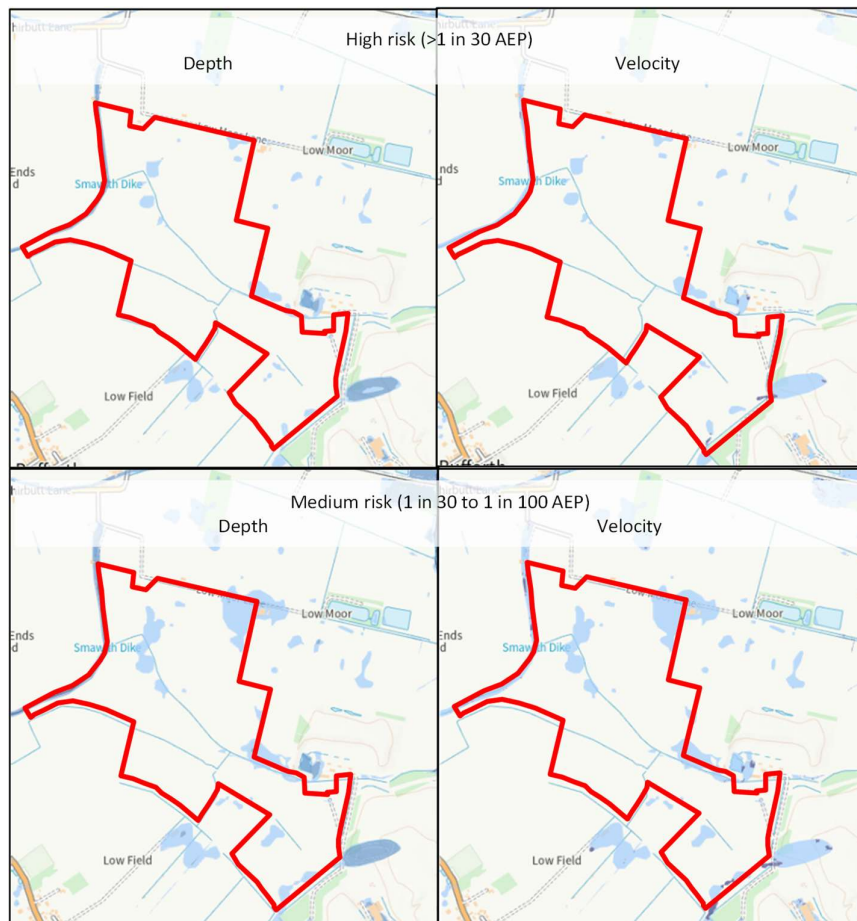
The zones do not take account of the possible impacts of climate change and consequent changes in the future probability of flooding. The Environment Agency has advised (refer consultation response provided in **Appendix C**) that detailed modelling is not available for this area, and as such, there is no modelled climate change data for this location.

4.3 Flood Risk from Small Watercourses and Surface Water (Pluvial)

As detailed in **Section 2.3**, a number of drainage ditches are located within the vicinity of the site. No modelled information is available for these watercourses. The Flood Risk from Surface Water map (**Figure 5**) has therefore been utilised to assess the risk of flooding from these sources.

This mapping indicates that during a high frequency/low severity (1 in 30 AEP) event, there is the potential for surface water ponding within isolated areas within the northern portion of the site and adjacent to the south-eastern boundary of the site, with flood depths indicated to be generally between 150 to 300 mm. During a medium risk event (1 in 30 to 1 in 100 AEP), flood extents are shown to increase and coincide with small sections of the on-site drainage ditches. During a low frequency/high severity (1 in 1,000 AEP) event, widespread flooding is indicated within the northern half of the site between Smawith Dyke and Low Moor Lane Dyke, with flood extents also increasing within the south-eastern portion of the site. Flood depths are shown to generally be between 150 to 300 mm, albeit depths are indicated to increase up to 600 mm in some localised areas within the northern half of the site.

The velocity during all events is indicated to generally be less than 0.25 m/s, owing to the flat topography of the land.



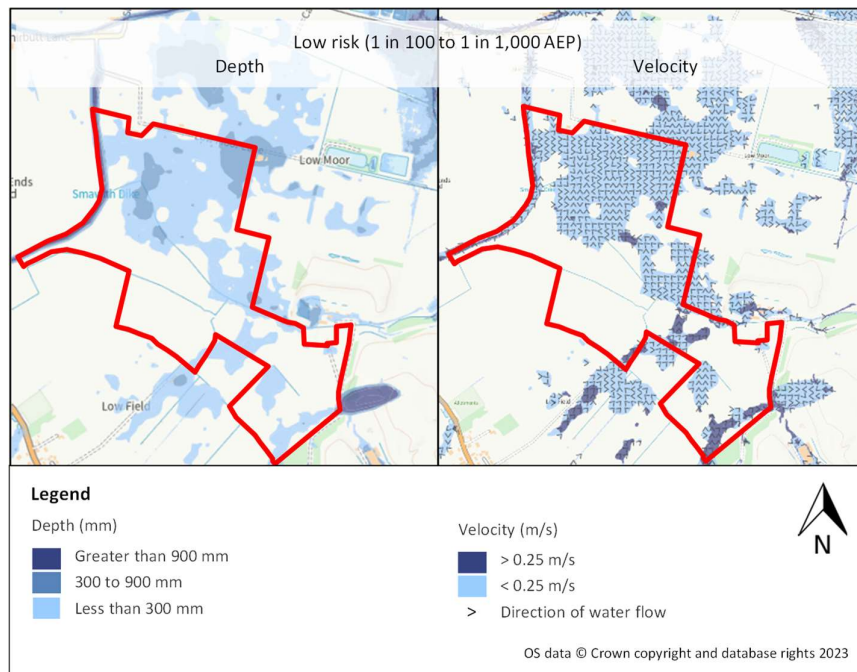


Figure 5: Flood Risk from Surface Water

Source: gov.uk website; Accessed: February 2023

4.4 Flood Risk from Reservoirs, Canals and Other Water Impounding Structures

There are no canals located within the immediate vicinity of the site. The Flood Risk from Reservoirs map (not shown) indicates that the site is not at risk of flooding from such sources. Land between Laybourne Lakes and the unnamed pond and the site is at higher elevations than the water impounded structures and the site.

It is concluded that the site is not at risk of flooding from reservoirs, canals or other water impounding structures.

4.5 Flood Risk from Groundwater

The JBA Groundwater Flood Risk Indicator map (not shown) indicates that the site is at a Negligible risk of groundwater flooding during a 1 in 100 AEP groundwater flood event.

4.6 Flood Risk Mitigation

The risk of flooding to the proposed development will be mitigated through the implementation of the following measures:

- Existing drainage ditches to be retained.
- No substations, inverter/transformer stations or storage containers located within 9 m of the top of bank of a board maintained watercourse, or 3 m of top of bank of an ordinary watercourse that is not Board maintained.
- Substations transformer stations and storage containers located outside of the 1 in 1,000 AEP surface water flood extent and at a minimum of 150 mm above adjacent ground levels.
- Substations, transformer stations and storage containers located within the 1 in 1,000 AEP surface water flood extent to be set at a minimum of 300 mm above adjacent ground levels .
- Substations, transformer stations and storage containers should not to be located within areas where flood depths are indicated to be greater than 300 mm within the 1 in 1,000 AEP surface water flood extent identified on the Flood Risk from Surface Water map or within flood zone 2 on the Flood Map for Planning.

- It is recommended that the area under the panel drip line is seeded with a suitable grass mix to prevent rilling and an increase in surface water runoff rates.
- Any new access crossings on the existing drainage ditches should be designed to maintain existing conveyance capacity.

5 SURFACE WATER MANAGEMENT

5.1 Surface Water Drainage at the Existing Site

As described in **Section 2.1**, the site currently comprises agricultural farmland. It is possible that field drains are present, but no other formal drainage infrastructure is believed to be present. Given ground conditions and site topography, surface water runoff would be expected to slowly infiltrate where conditions allow and flow overland in a direction determined by topography.

The greenfield surface water runoff rates for the site, calculated using the ICP SUDS method within MicroDrainage are presented in **Table 1**. Details of the input parameters and the output results are provided in **Appendix D**.

Table 1: Greenfield Runoff Rate

AEP of Rainfall Event	Greenfield Runoff Rate (l/s/ha)	Greenfield Runoff Rate for 61.1 ha Site (l/s)
1 in 1	3.4	207.7
QBAR	4.0	244.4
1 in 30	7.0	427.7
1 in 100	8.2	501.0

5.2 Surface Water Drainage at the Developed Site

Panelled Areas of the Solar Farm

According to published research into the impact of solar-farm panels on runoff rates and volumes⁵, solar panels do not have a significant impact on the hydrologic response of a site when the ground comprises of well managed vegetation such as good grass cover. In such instances, the research cites that well managed vegetation beneath the solar panels would limit the potential increase in runoff volume to less than 0.35%.

Given that the site is virtually flat, and that the proposed development includes managed grassland beneath the solar panels, the impact on runoff rates and volumes from the panelled part of the development is assessed to be negligible. As such, no specific drainage for the panelled part of the site is proposed.

Access Tracks and Other Areas of Hardstanding

In accordance with PPG para. 056, surface water runoff should be disposed of according to the following hierarchy: Into the ground (infiltration); To a surface water body; To a surface water sewer, highway drain, or another drainage system; To a combined sewer.

As detailed in **Section 2.4**, the site is underlain by soils with impeded drainage, overlying clay. Although infiltration testing has not been undertaken at this stage, the desktop study, together with the density of the network of agricultural drainage ditches, provides strong evidence that the disposal of surface water via infiltration is unlikely to be feasible. It is therefore proposed to direct runoff from the developed site to the on-site drainage ditches.

For the purposes of this assessment, the site has been subdivided into four drainage areas based on the location of surface waterbodies, as illustrated on **Figure 6**.

As outlined in **Section 3.3**, Ainsty (2008) IDB advise that Type 3 material is classed as being permeable. It is therefore recommended that Type 3 material is used for the construction of the access tracks. Where runoff generated by the access tracks cannot infiltrate, outfalls located across the site will direct runoff to the on-site drainage ditches. For the purposes of this assessment, the access tracks are considered to be impermeable to represent a worst case scenario.

The contributing areas within each drainage area associated with the access tracks, transformer stations, sub-stations and storage containers, has been calculated based on the proposed site layout plan and is presented in **Table 2**.

⁵ 'Hydrologic Response of Solar Farms', Cook LM and McCuen RH, American Society of Civil Engineers, 2013

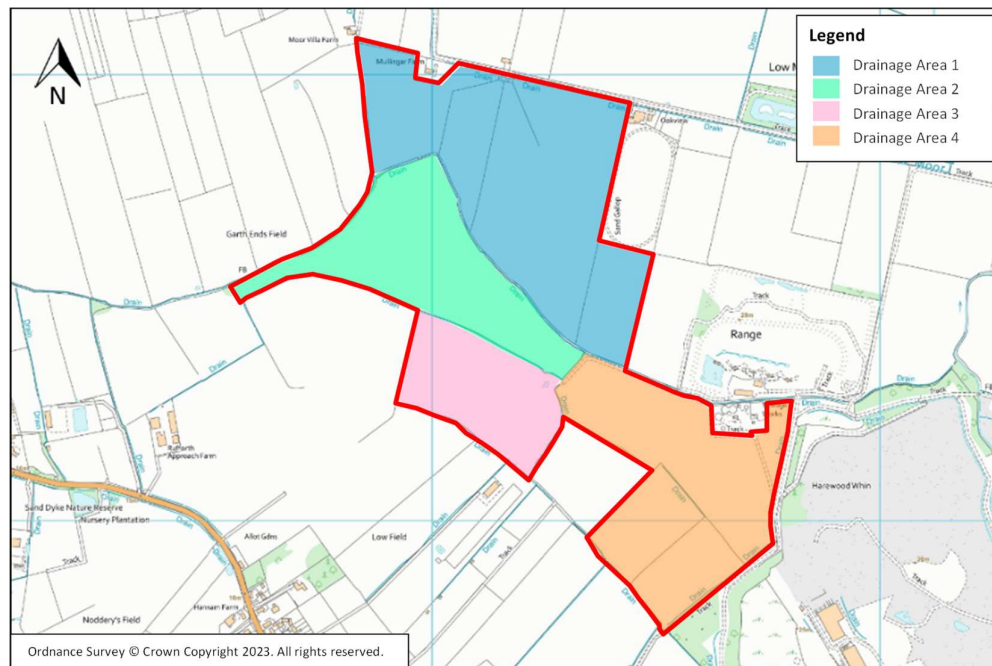


Figure 6: Indicative Drainage Areas

5.2.1 Peak Flow Control

Ainsty (2008) IDB requires surface water runoff to be restricted to 1.4 l/s/ha. This runoff rate will be achieved so far as is practicable; however, it is recognised that a flow control with a diameter of less than 50 mm may pose a risk of blockage to the drainage system. The proposed peak discharge for each drainage area is presented in **Table 2**.

5.2.2 Volume Control

As outlined within the CIRIA SuDS Manual 2015 extra runoff volumes in extreme events may be managed by releasing all runoff (above the 1 in 1 AEP event) from the site at a maximum rate of 2 l/s/ha or QBAR, whichever is the higher value.

It is therefore proposed to restrict peak discharge rates to 1.4 l/s/ha recognising a flow control with a minimum diameter of 50 mm would apply in up to the 1 in 100 AEP event, including an allowance for climate change.

5.2.3 Attenuation Storage

Attenuation storage will be provided to store surface water runoff generated across contributing areas.

The attenuation storage facilities have been modelled using the Source Control module of MicroDrainage (**Appendix E**). The required storage volume has been sized to store the 1 in 100 AEP rainfall event including a 45% increase in rainfall intensity to allow for climate change in accordance with Environment Agency guidance⁶. The use of a 45% allowance for climate change is greater than the 30% guidance required by Ainsty (2008) IDB.

The proposed attenuation volumes are provided in **Table 2** below.

⁶ Flood Risk Assessments: climate change allowances (<https://www.gov.uk/guidance/flood-risk-assessments-climate-change-allowances>)

Table 2: Summary of Proposed Surface Water Drainage Scheme
(1 in 100 AEP Event plus 45% Climate Change)

Drainage Area	Outfall	Contributing Area (ha)	Peak Discharge (l/s)	Attenuation Volume (m ³)
1	S1a	0.165	0.2	98.7
	S1b	0.319	0.5	222.4
2	S2	0.358	0.5	259.4
3	S3	0.148	0.2	86.2
4	S4a	0.162	0.2	96.5
	S4b	0.234	0.3	152.4

*Based on a flow control (e.g. Hydro-Brake) diameter of 50 mm

The storage volumes could be accommodated within a Type 3 sub-base material within the access tracks at a depth of 0.3 m, apart from the access road serving outfall S2 which would require a sub-base depth of 0.4 m.

Should a gravity system not be feasible, a pumped solution would be required. The minimum pump rate achievable is 2.0 l/s, subject to site specific drainage parameters. This will be determined at the detailed design stage.

An indicative surface water drainage layout is provided in **Appendix A**.

5.2.4 Pollution Control

Table 26.2 of the CIRIA SuDS Manual (2015) identifies general access roads as having a low pollution hazard level and indicates that the pollution hazard indices associated with general access roads for total suspended solids, hydrocarbons and metals are 0.50, 0.40 and 0.40 respectively.

It is recommended that the access tracks are constructed with a Type 3 material to allow surface water runoff to be stored and treated within the pavement structure.

Table 26.3 of the CIRIA SuDS Manual indicates that the SuDS mitigation indices for permeable paving for total suspended solids, hydrocarbons and metals are 0.70, 0.60 and 0.70 respectively.

The CIRIA SuDS manual identifies commercial roofs (inverter/transformer stations, sub-stations and storage containers) as having a low pollution hazard level, as such minimal treatment measures would be required to treat the quality of the water.

5.2.5 Adoption and Maintenance of SuDS

The surface water drainage system will be maintained by the site operator or their appointed management company. An indicative maintenance schedule is presented in **Table 3**.

Table 3: Maintenance Requirements

Schedule	Required action	Frequency
Type 3 Permeable Access Tracks		
Occasional maintenance	Stabilise and mow contributing and adjacent areas	As required
	Removal of weeds or management using glyphosphate applied directly into the weeds by an applicator rather than spraying	As required – once per year on less frequently used pavements
Remedial actions	Remediate any landscaping which, through vegetation maintenance or soil slip, has been raised to within 50mm of the level of the paving	As required
	Remedial work to any depressions and rutting considered detrimental to the structural performance or a hazard to users	

Schedule	Required action	Frequency
	Rehabilitation of surface and substructure	Every 10 to 15 years or as required
Monitoring	Initial inspection	Monthly for three months after installation
	Inspect for evidence of poor operation and/or weed growth- if required, take remedial action	Three-monthly, 48h after large storms in first six months
	Inspect silt accumulation rates and establish appropriate removal frequencies	Annually
Flow Control Unit		
Routine maintenance	Remove litter and debris and inspect for sediment accumulation	Six Monthly
	Remove sediment from sump	As necessary – Indicated by system inspections
Remedial actions	Replace malfunctioning parts or structures	As required
Monitoring	Inspect for evidence of poor operation	Six Monthly
	Inspect flow control unit and establish appropriate replacement frequencies	Six Monthly
	Inspect sediment accumulation rates and establish appropriate removal frequencies	Monthly during first year of operation, then every six months

6 SUMMARY AND RECOMMENDATIONS

This report has been prepared for Atmos Consulting on behalf of Hessay Solar Limited and relates to the proposed development of land at Hessay, York for use as a solar farm.

The Environment Agency Flood Map for Planning indicates the site to be predominantly located in flood zone 1, with the north-eastern corner of the site located in flood zone 2. The north-western boundary of the site is indicated to abut flood zones 3 and 2.

The site is assessed to be at a low to medium risk of flooding from small watercourses and surface water. During a low frequency/high severity (1 in 1,000 AEP) event, there is the potential for widespread flooding within the northern half of the site between Smawith Dyke and Low Moor Lane Dyke, and within the south-eastern portion of the site, with flood depths generally between 150 to 300 mm. Flood extents are shown to significantly reduce during less extreme events.

It is assessed that the site is not at risk of flooding from groundwater, reservoirs, canals or other water impounding structures.

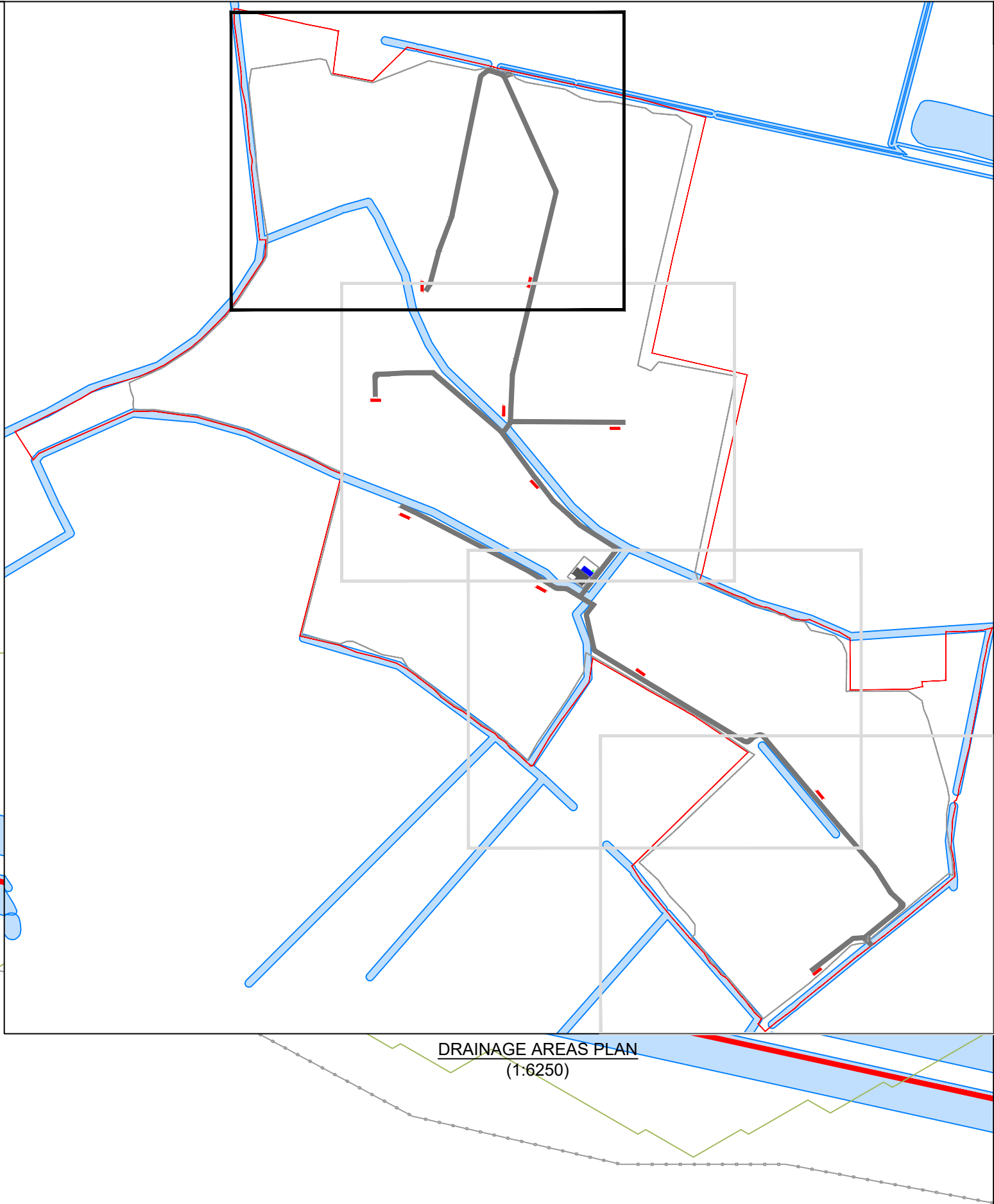
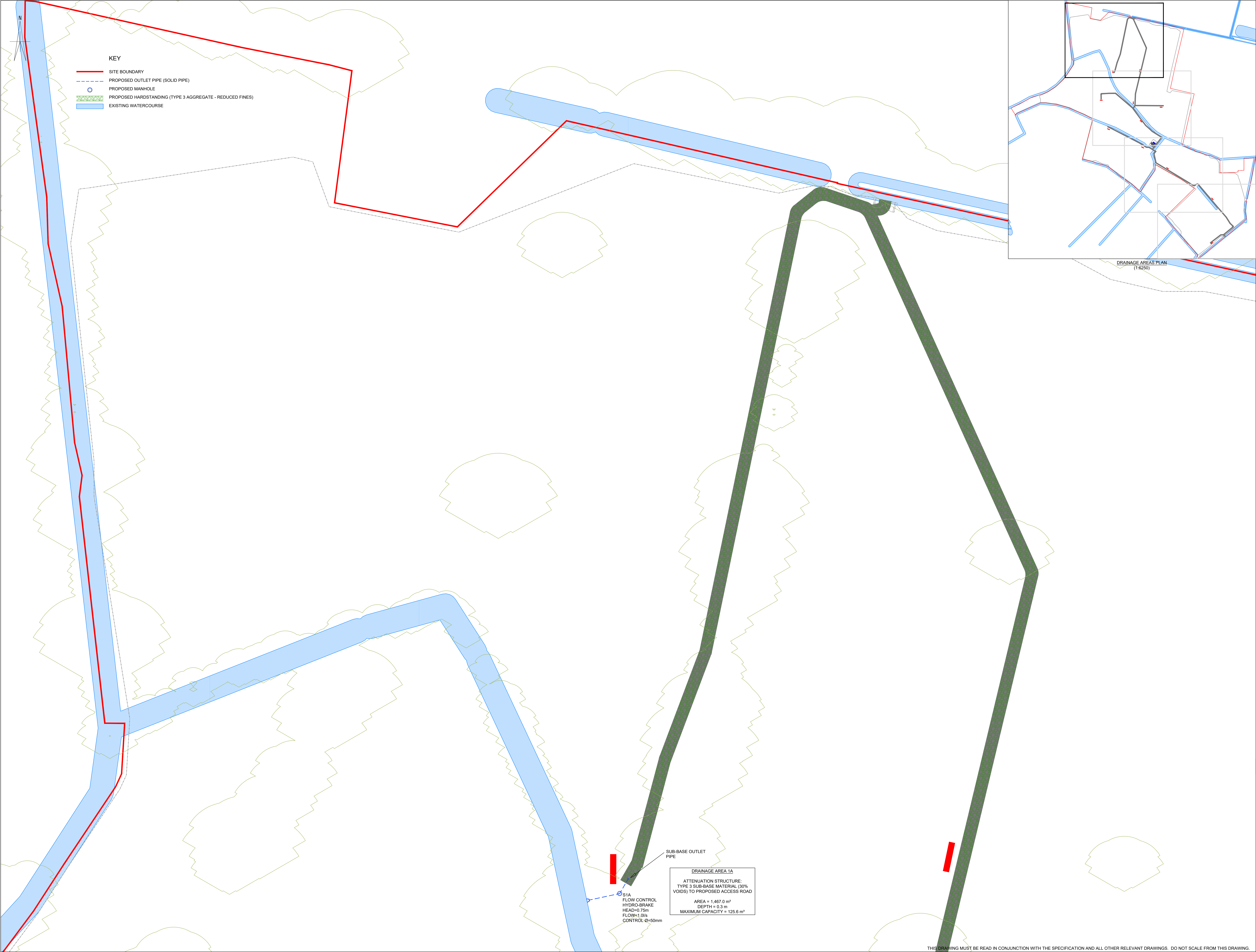
The assessment demonstrates that the proposed development may be completed in accordance with the requirements of planning policy subject to the following:

- Existing drainage ditches to be retained.
- No substations, inverter/transformer stations or storage containers located within 9 m of the top of bank of a board maintained watercourse, or 3 m of top of bank of an ordinary watercourse that is not Board maintained.
- Substations transformer stations and storage containers located outside of the 1 in 1,000 AEP surface water flood extent and at a minimum of 150 mm above adjacent ground levels.
- Substations, transformer stations and storage containers located within the 1 in 1,000 AEP surface water flood extent to be set at a minimum of 300 mm above adjacent ground levels .
- Substations, transformer stations and storage containers should not to be located within areas where flood depths are indicated to be greater than 300 mm within the 1 in 1,000 AEP surface water flood extent identified on the Flood Risk from Surface Water map or within flood zone 2 on the Flood Map for Planning.
- Panelled part of the site to comprise managed grassland.
- Ground under the solar panel drip line to be seeded with a suitable grass mix to prevent rilling.
- Any new access crossings on existing drainage ditches to be designed to maintain existing conveyance capacity.
- The proposed maintenance access track to be constructed from a Type 3 permeable pavement aggregate to promote water quality treatment.
- Surface water runoff from the access tracks and other areas of hardstanding to be restricted to flow rates to suit a flow control diameter of 50 mm as to avoid the risk of blockages, with storage provided within the Type 3 access track material.

In conclusion, this report demonstrates that the proposed development may be completed in accordance with the requirements of planning policy.

APPENDIX A

Indicative Drainage Layout



NOTES

- THIS DRAWING TO BE READ IN CONJUNCTION WITH ALL RELEVANT WEETWOOD DRAWINGS.
- THE PROPOSED SITE LAYOUT IS INDICATIVE ONLY.

P2	21.03.23	SOLAR PANELS OMITTED	TB	KT
P1	24.02.23	INITIAL ISSUE	TB	KT

REV	DATE	DESCRIPTION	DRAWN	CHECK

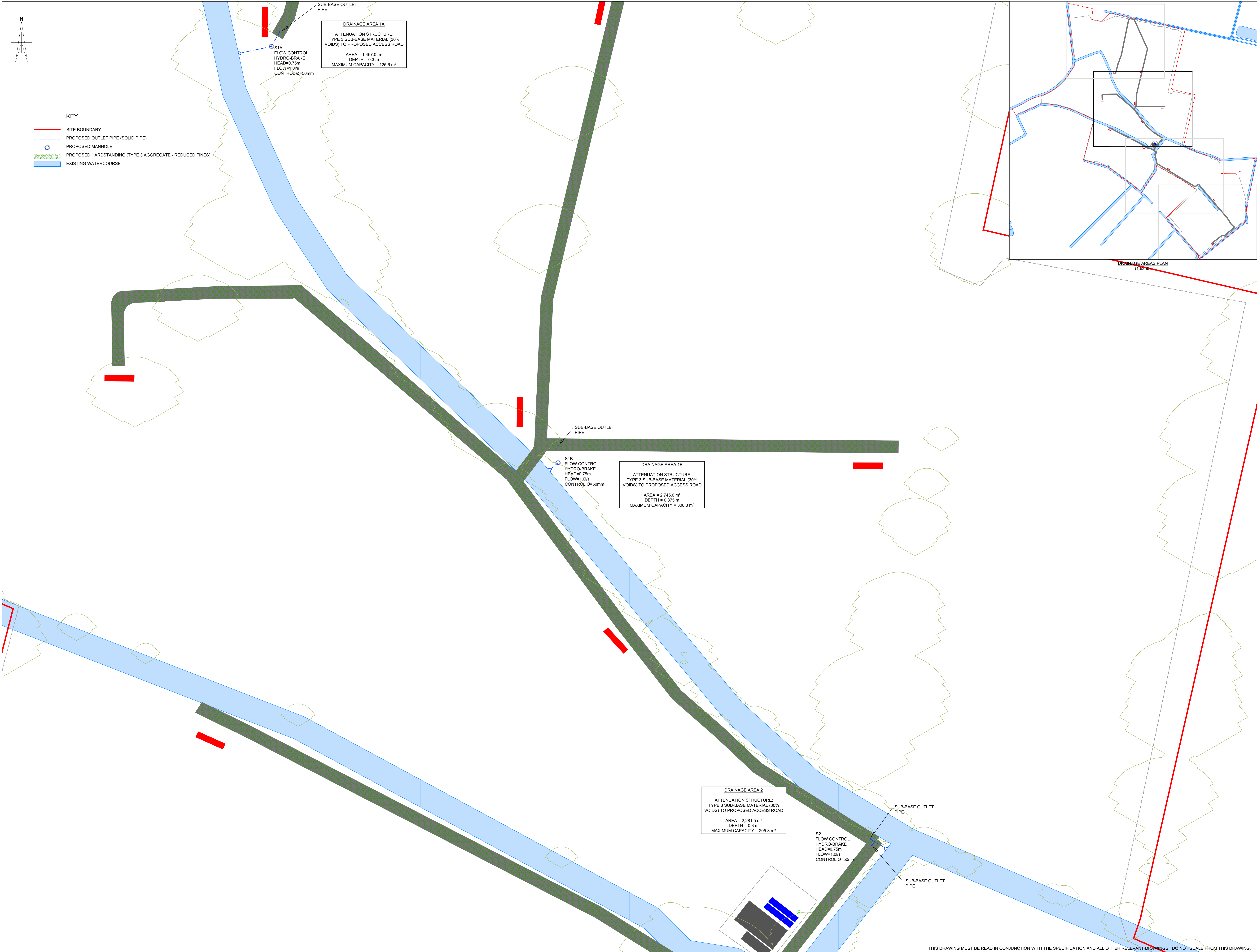
Development • Planning • Environment

Park House, Ffordd
Byrnw Gwair, Mold
CH7 1FQ
Tel 01352 700045
info@weetwood.net
www.weetwood.net

Client

HESSAY SOLAR LIMITED

Drawing Status	Date
PRELIMINARY	FEB 2023
Project	Scale (A0)
LAND AT HESSAY YORK	1:500
Project No	Drawn
5767	TB
	Checked
	KT
Title	Drawing No
INDICATIVE SURFACE WATER DRAINAGE LAYOUT 1 OF 4	101
	Revision
	P2



NOTES

1. THIS DRAWING TO BE READ IN CONJUNCTION WITH ALL RELEVANT WEETWOOD DRAWINGS.

2. THE PROPOSED SITE LAYOUT IS INDICATIVE ONLY.

P2	21.03.23	SOLAR PANELS OMITTED	TB	KT
P1	24.02.23	INITIAL ISSUE	TB	KT

REV	DATE	DESCRIPTION	DRAWN	CHECK

Client

HESSAY SOLAR LIMITED

Drawing Status

PRELIMINARY

Project

LAND AT HESSAY YORK

Title

INDICATIVE SURFACE WATER DRAINAGE LAYOUT
2 OF 4

Date

FEB 2023

Scale (A0)

1:500

Drawn

TB

Checked

KT

Project No

5767

Drawing No

102

Revision

P2

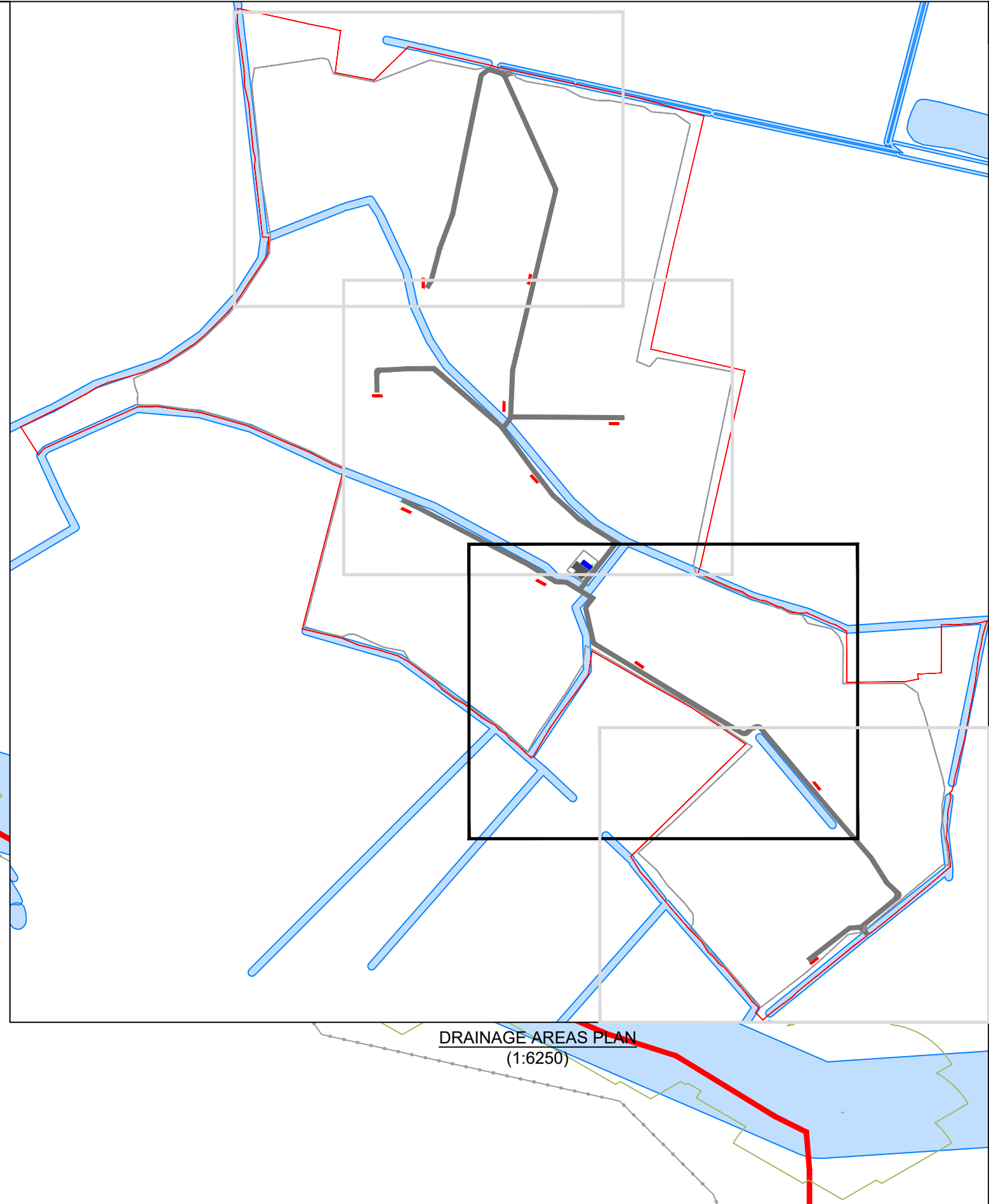
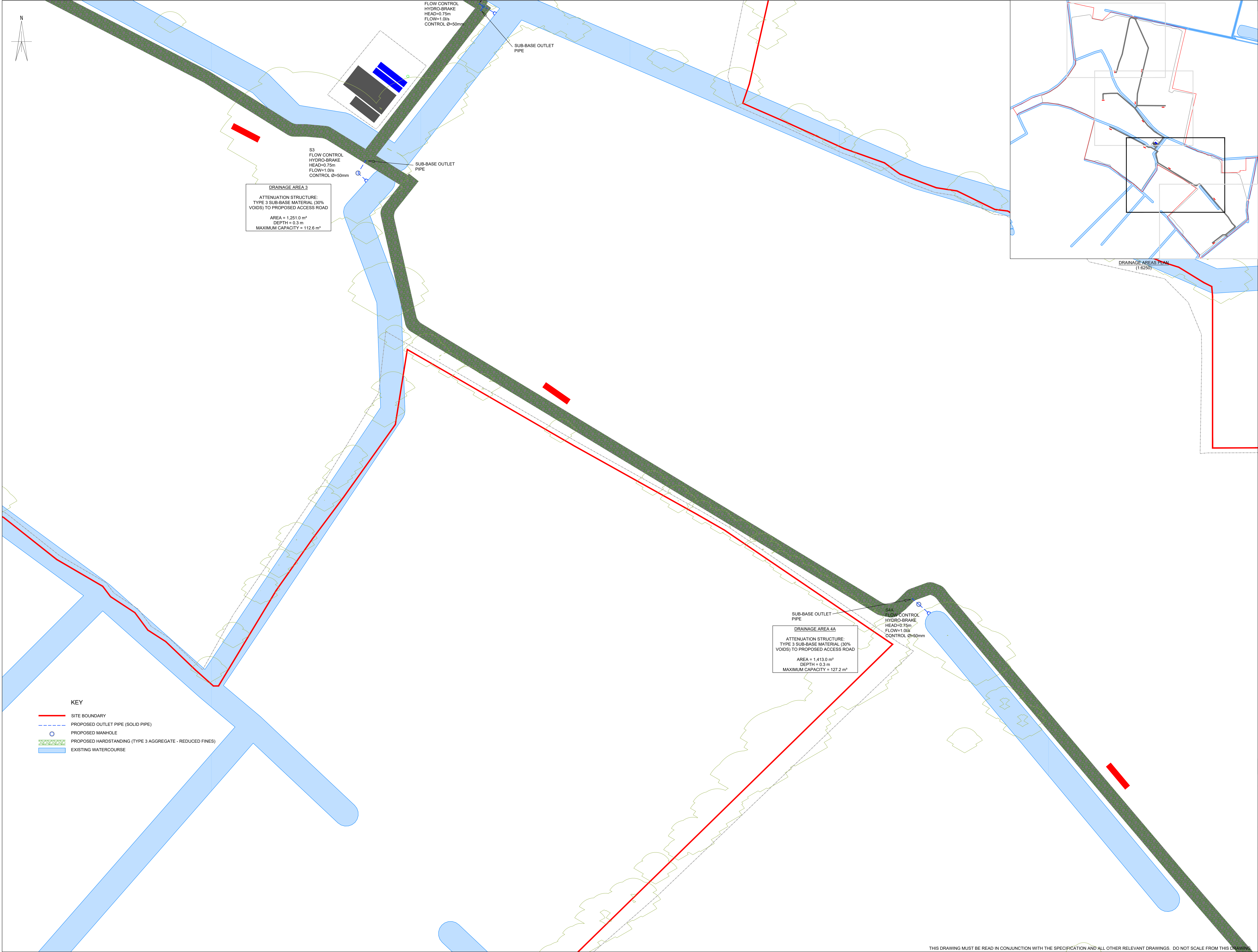
Park House, Ffordd Bymwr Gwair, Mold CH7 1FQ
Tel 01352 700045
info@weetwood.net
www.weetwood.net

Client

Development • Planning • Environment

Drawn

THIS DRAWING MUST BE READ IN CONJUNCTION WITH THE SPECIFICATION AND ALL OTHER RELEVANT DRAWINGS. DO NOT SCALE FROM THIS DRAWING.



NOTES

1. THIS DRAWING TO BE READ IN CONJUNCTION WITH ALL RELEVANT WEETWOOD DRAWINGS.

2. THE PROPOSED SITE LAYOUT IS INDICATIVE ONLY.

P2	21.03.23	SOLAR PANELS OMITTED	TB	KT
P1	24.02.23	INITIAL ISSUE	TB	KT

REV	DATE	DESCRIPTION	DRAWN	CHECK

Client

HESSAY SOLAR LIMITED

Drawing Status

PRELIMINARY

Project

LAND AT HESSAY YORK

Project No

5767

Title

INDICATIVE SURFACE WATER DRAINAGE LAYOUT

3 OF 4

Date

FEB 2023

Scale (A0)

1:500

Drawn

TB

Checked

KT

Project No

5767

Drawing No

103

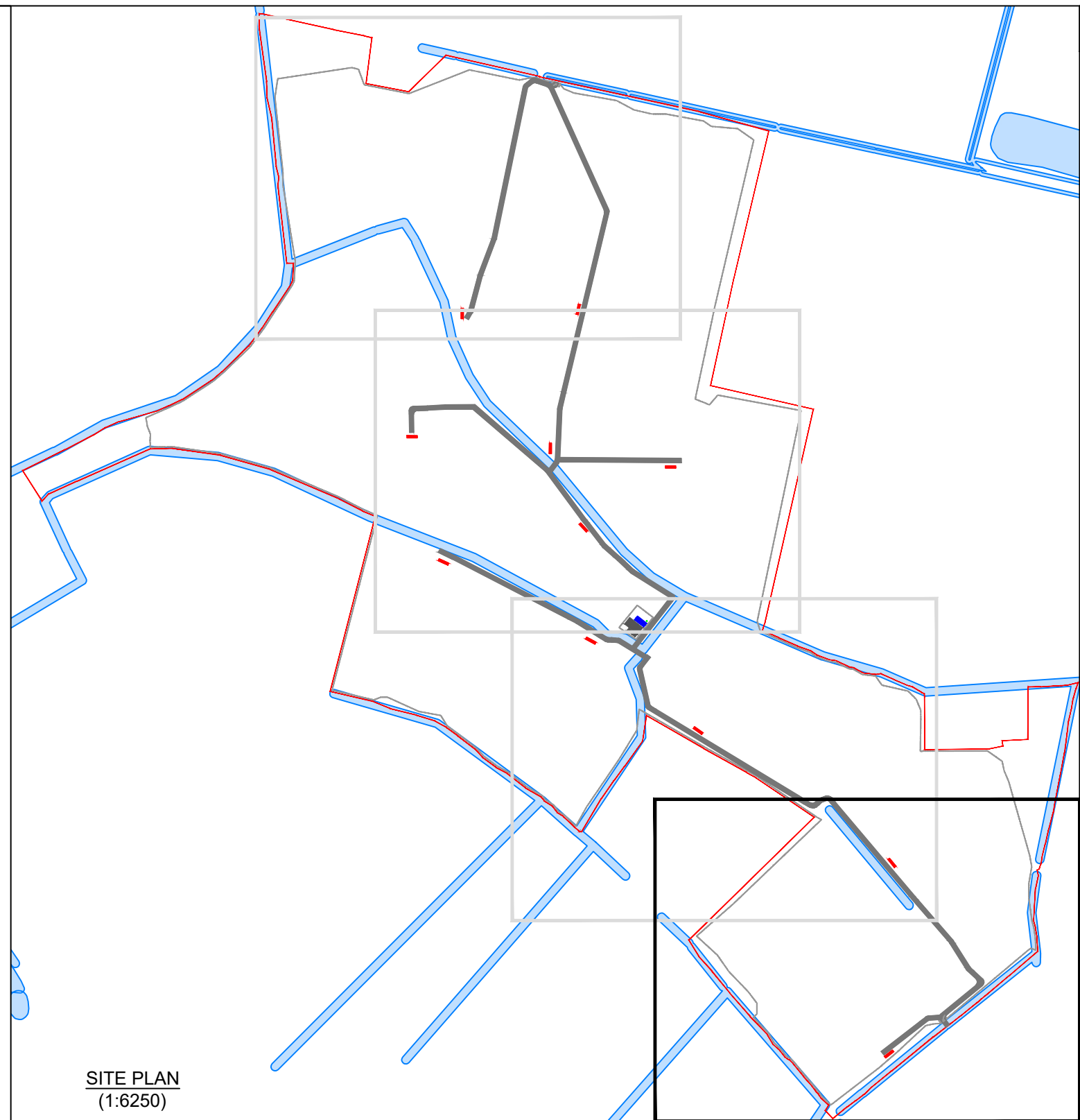
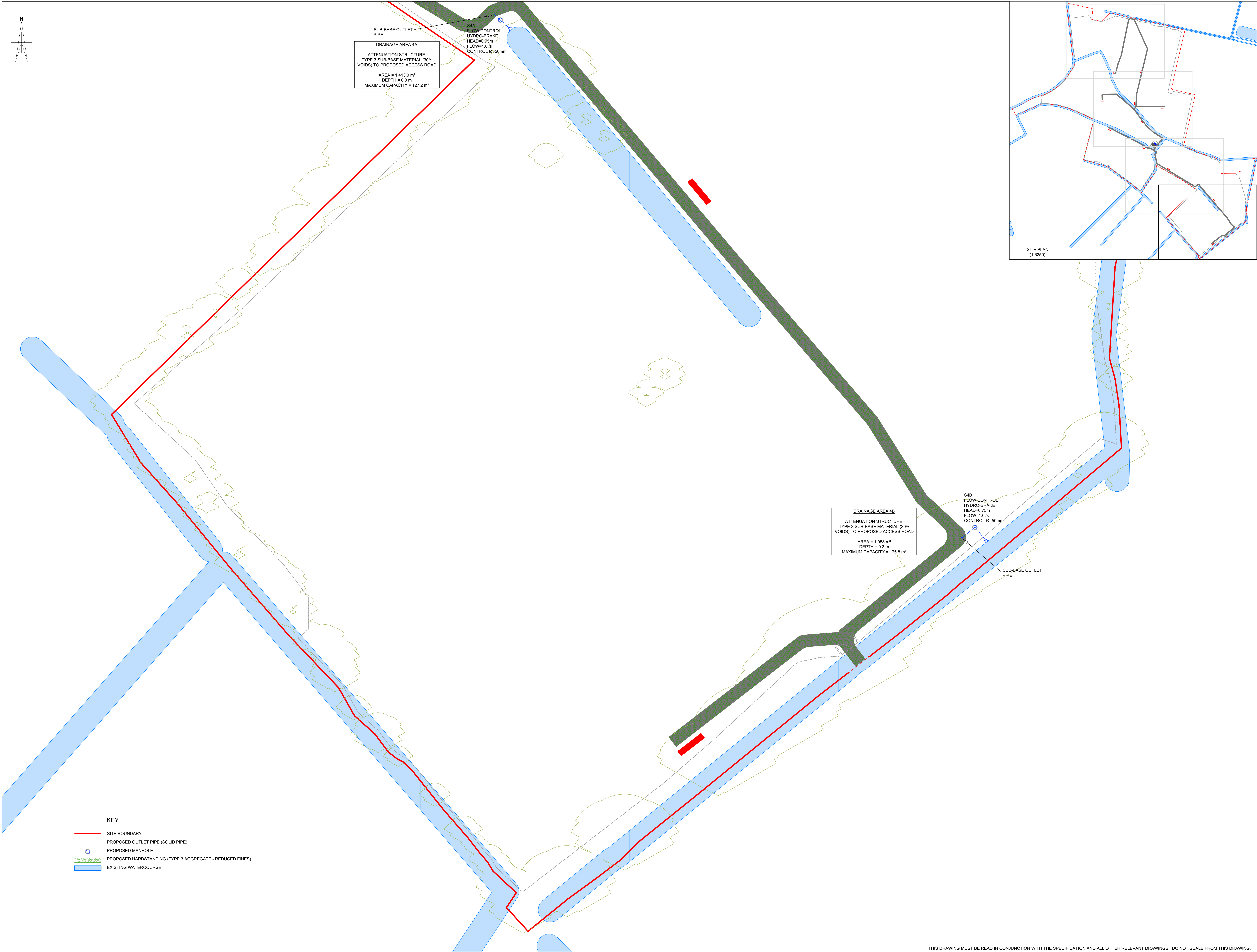
Revision

P2

Park House, Fford
Bynmwr Gwair, Mold
CH7 1FQ
Tel 01352 700045
info@weetwood.net
www.weetwood.net

Development • Planning • Environment

THIS DRAWING MUST BE READ IN CONJUNCTION WITH THE SPECIFICATION AND ALL OTHER RELEVANT DRAWINGS. DO NOT SCALE FROM THIS DRAWING.



NOTES

- THIS DRAWING TO BE READ IN CONJUNCTION WITH ALL RELEVANT WEETWOOD DRAWINGS.
- THE PROPOSED SITE LAYOUT IS INDICATIVE ONLY.

P2	21.03.23	SOLAR PANELS OMITTED	TB	KT
P1	24.02.23	INITIAL ISSUE	TB	KT
REV	DATE	DESCRIPTION	DRAWN CHECK	

Weetwood
Development • Planning • Environment

Park House, Ffordd
Byrnwr Gwair, Mold
CH7 1FQ
Tel 01352 700045
info@weetwood.net
www.weetwood.net

Client

HESSAY SOLAR LIMITED

Drawing Status	Date
PRELIMINARY	FEB 2023
Project	Scale (A0)
LAND AT HESSAY YORK	1:500
Project No	Drawn
5767	TB
	Checked
	KT
Title	Project No
INDICATIVE SURFACE WATER DRAINAGE LAYOUT	5767
4 OF 4	Drawing No
	104
	Revision
	P2

APPENDIX B

Ainsty (2008) Internal Drainage Board Consultation Response

From: [Planning](#)
To: [Keely Bonser](#)
Cc: [Nathan Culpan](#); [Bill Symons](#)
Subject: RE: 5767 Low Moor Lane, Hessay, York - Flood Risk Information Request [Filed 28 Nov 2022 13:00]
Date: 28 November 2022 12:26:39
Attachments: [image001.jpg](#)
[image002.png](#)
[image003.png](#)
[image004.jpg](#)
[image005.png](#)
[image006.png](#)

Dear Keely,

Further to my below e-mail, our Engineer has now confirmed:

“The Smawwith dyke 080 – is known to overspill onto the land adjacent and store a vast amount of water leading up to Mullingar farm house to the north. Flood water remains on the land for a short amount of time until levels/rainfall recede enough- Smawwith dyke is able to run both to the east and west outfalling into different watercourses that ultimate lead to the river Ouse. If a bund was to be installed then the watercourse would not have the ability to overtop / store water and this could push flood water elsewhere.”

This is just general knowledge and should not replace formal flood investigations.

Kind regards,

Charlotte Gill

Planning Officer

Please note my new working hours are Monday, Wednesday, Thursday – 8.45am to 4.30pm.



Address: Derwent House | Crockey Hill | York | YO19 4SR

E-mail: planning@yorkconsort.gov.uk

Telephone: 01904 720 785 (Option 4)

Website: <http://www.yorkconsort.gov.uk>



please don't print this e-mail unless you really need to

This e-mail, together with any attachments, is confidential and is intended solely for the individual to whom it is addressed. The views expressed are that of the author and do not constitute or imply the endorsement or recommendation of the Drainage Board. If you are not the intended recipient please inform the sender immediately.

Whilst the Board does run anti-virus software, you are solely responsible for ensuring that any e-mail or attachment you receive is virus free and the Board disclaims any liability for any damage you suffer as a consequence of receiving any virus.

From: Planning

Sent: 24 November 2022 10:42

To: 'Keely.Bonser@weetwood.net' <Keely.Bonser@weetwood.net>

Cc: Bill Symons <Bill.Symons@yorkconsort.gov.uk>; Nathan Culpan
<nathan.culpan@yorkconsort.gov.uk>

Subject: RE: 5767 Low Moor Lane, Hessay, York - Flood Risk Information Request

Hello Keely,

Bill has forwarded me the below e-mail because I deal with any applications for our Ainsty (2008) Internal Drainage Board.

-

Generally

I have attached a map showing the location of the Board's district in green. Board maintained watercourses are shown in blue.

I have also attached a plan I have roughly marked up, showing the location of our Board maintained watercourses within your red line boundary.

There may also be ordinary watercourses nearby which are not maintained by the Board and we assume remain with the riparian owner to maintain. However, the Board's consent is still required in certain instances set out below.

Board's Consent

-

Under the Land Drainage Act 1991 and the Boards' byelaws, the Board's **prior written** consent (outside of the planning process) is needed for:-

- a. any connection into a Board maintained watercourse, or any ordinary watercourse in the Board's district.
- b. any discharge, or change in the rate of discharge, into a Board maintained watercourse, or any ordinary watercourse in the Board's district. ***This applies whether the discharge enters the watercourse either directly or indirectly (i.e. via a third party asset such as a mains sewer).***
- c. works within or over a Board maintained watercourse, or any ordinary watercourse in the Board's district – for example, land drainage, an outfall structure, bridges, culverting etc.
- d. any construction, fencing or planting within 9 metres of the top of the embankment of a Board maintained watercourse (as shown on the below diagram)



Please note that the Board does not, generally, own any watercourses and the requirement for you to obtain the Board's consent is in addition to you obtaining consent from any land owner or other authority to carry out the relevant works.

- Full details of the Consent process can be found on our website:- <http://www.yorkconsort.gov.uk>

Works very close to Board maintained watercourses

- The Board's prior written consent is required for any new structures or fences or planting or hardstanding within 9 metres of the top of the embankment of any Board maintained watercourse.

The Board would usually resist any such new structures as we must have sufficient access to these watercourses to maintain them.

In addition, we want to ensure that there is nothing to destabilise the embankments of the watercourses.

Works very close to "ordinary watercourses" not maintained by the Board

- For any watercourses not maintained by the Board, we would be considering what works are proposed over or very close to them.

We would usually state a minimum of 3 metres from the top of the embankment of any ordinary watercourse is left completely clear.

Again, we would want to ensure that there is nothing that could destabilise the embankments over time and effect the flows of the watercourses.

New Access Tracks over Watercourses

-

If you are proposing to create any access tracks over a watercourse within the Board's district, the Board's prior written consent must be obtained.

We would request full details of the proposed bridges and want to ensure that any culvert is sufficiently sized for the watercourse so there is no effect on the water flows.

Underground Cables

-

If you are proposing to insert a cable under a watercourse within the Board's district, the Board's prior written consent must again be obtained. In these instances, we would prefer Directional Drilling to be used.

-

Surface Water

The Board would view any solar panels as being an "impermeable area".

Please note that we also view Type 1 material as impermeable if you are considering this for any access roads – Type 3 would however be seen as permeable.

We would therefore usually seek:

1. Soakaways

The Board always recommends that soakaways are *first* considered in accordance with the Planning Practice Guidance hierarchy for the management of surface water. The Board would therefore recommend:-

i. Percolation Testing

That the applicant be asked to carry out soakaway testing, in accordance with BRE Digest 365, in order to ascertain whether the soil structure is suitable for a soakaway system.

ii. Soakaway Design

Should the testing prove to be successful the applicant should then submit a design for the soakaway, for approval by the Lead Local Flood Authority ("LLFA") as the "approving authority" for soakaways, which should:-

- i. Storage volume should accommodate a 1:30 year event with no surface flooding (plus 30% allowance for climate change); and
- ii. Storage volume should accommodate no overland discharge off the site in a 1:100 year event (plus 30% allowance for climate change).

Even if a soakaway already exists, the Board would suggest that the applicant provides confirmation of its location and that the system is working effectively, and also have evidence that it is capable of handling the volume of water that will be generated by the development. It is not, usually, sufficient for the applicant to rely on anecdotal evidence of its past performance.

2. Discharge into a Watercourse

The Board will only accept a discharge into a watercourse (directly or indirectly) where soakaways are not feasible. The below requirements apply when:

- There is a direct discharge to a watercourse.
- There is an indirect discharge to a watercourse – for example, through a private drainage system, or a mains sewer, which eventually discharges into a watercourse.

a. Details of the Watercourse / Sewer

The Board would request details of:

- i. What the applicant is proposing to discharge into – for example, a watercourse.
- ii. The location of the proposed point of connection.

b. Flow of the Watercourse

If the applicant is proposing to discharge directly (or through private drainage) into a watercourse, and if that watercourse is not maintained by the Board, we would ask:

- i. Where this watercourse is flowing to. A simple plan showing the route of the watercourse to the nearest Board maintained watercourse is usually sufficient.
- ii. Details of the condition of the watercourse to ensure the same is flowing freely prior to any discharge. The applicant is responsible for ensuring that the watercourse is free flowing but we would ask that they walk along the watercourse and ensure there are no blockages. Photographs should be provided as evidence.

c. Discharge Rate

The amount of water should be restricted to an agreed rate, using the below requirements:

- i. The applicant should first demonstrate that there is an existing operational connection to the watercourse for the development site. This should be done by way of Dye Testing or a CCTV Survey.

Where that connection is established, the Board would want to know the size of those existing connected impermeable areas.

The existing drainage rate should then be calculated as 140 litres per second per hectare for the connected impermeable area, or the established rate (whichever is the lesser) - less 30%.

The applicant may also add an amount for any new areas of the site which will now be positively drained (but which were not positively drained before) – this should be at the “greenfield” rate of 1.4 litres per second per hectare.

- ii. Where there is a new connection to a watercourse or to a sewer that discharges to a watercourse, the maximum discharge that will be accepted is at

the “greenfield” rate of 1.4 litres per second per hectare.

- iii. If the site has been lying vacant and/or demolished before the existing surface water discharge regime is determined, then the maximum discharge that will be accepted from an area that is shown to discharge to the watercourse is greenfield run-off rates.

d. Flow Control Device

Whilst the Board is not the “approving authority” for flow control devices, we would request simple details as to what is proposed with regards to how the flow will be restricted to the agreed discharge rate.

e. Surface Water Storage System

Again, the Board is not the “approving authority” for surface water storage systems. However, we would request details of:

- i. The proposed surface water storage system (which we would usually recommend is impermeably lined); and
- ii. The proposed storage volume and accompanying calculations.

The system should accommodate a 1:30 year event with no surface flooding (plus 30% allowance for climate change); and no overland discharge off the site in a 1:100 year event (plus 30% allowance for climate change).

We would however recommend that a system should try and accommodate the full 1:100 year storm event (plus 30% allowance for climate change) wherever possible.

f. Outfall Structure

If there is a direct discharge to a watercourse and if that watercourse is within our district, the applicant should also provide details of the proposed outfall structure into the watercourse.

Flood Data

-

We do not have any formal flood data we can share but I will ask our Engineer if he has any general information he can provide.

I hope this assists but if you require anything further, please let me know.

Kind regards,

Charlotte Gill

Planning Officer

Please note my new working hours are Monday, Wednesday, Thursday – 8.45am to 4.30pm.



Address: Derwent House | Crockey Hill | York | YO19 4SR

E-mail: planning@yorkconsort.gov.uk

Telephone: 01904 720 785 (Option 4)

Website: <http://www.yorkconsort.gov.uk>



please don't print this e-mail unless you really need to

This e-mail, together with any attachments, is confidential and is intended solely for the individual to whom it is addressed. The views expressed are that of the author and do not constitute or imply the endorsement or recommendation of the Drainage Board. If you are not the intended recipient please inform the sender immediately.

Whilst the Board does run anti-virus software, you are solely responsible for ensuring that any e-mail or attachment you receive is virus free and the Board disclaims any liability for any damage you suffer as a consequence of receiving any virus.

From: Keely Bonser <Keely.Bonser@weetwood.net>

Sent: 23 November 2022 12:15

To: Bill Symons <Bill.Symons@yorkconsort.gov.uk>

Subject: 5767 Low Moor Lane, Hessay, York - Flood Risk Information Request

Hi Bill

We're currently undertaking a flood risk assessment for a proposed solar farm in Hessay (see attached location map). A number of IDB owned and maintained watercourses cross the site (as listed below). Could you kindly provide any flood risk information you may hold on these watercourses (and any other relevant drains that you may hold data on) including historical flood records, flood levels etc.

78: Low Moor Lane Drain

80: Smawith Dyke

82: Oak Nursery Dyke

83: Rufforth North Dyke (B)

Also, do you have any requirements with regards to surface water runoff from solar farms, in particular runoff from solar panels?

Kind Regards

Keely

Keely Bonser

Director

Please note my working days are Monday to Thursday.



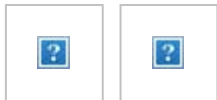
keely.bonser@weetwood.net

Suite C22 Joseph's Well, Hanover Walk, Leeds LS3 1AB

T 0113 244 1377 **M** 07745 241291

Offices also at:

Park House, Ffordd Byrnwr Gwair, Mold, CH7 1FQ (**T** 01352 700045)



[Click here to visit our company email disclaimer and policy documents](#)

We're recruiting! Please visit our [website](#) for further information

APPENDIX C

Environment Agency Consultation Response - Flood Risk Information

Flood risk assessment data

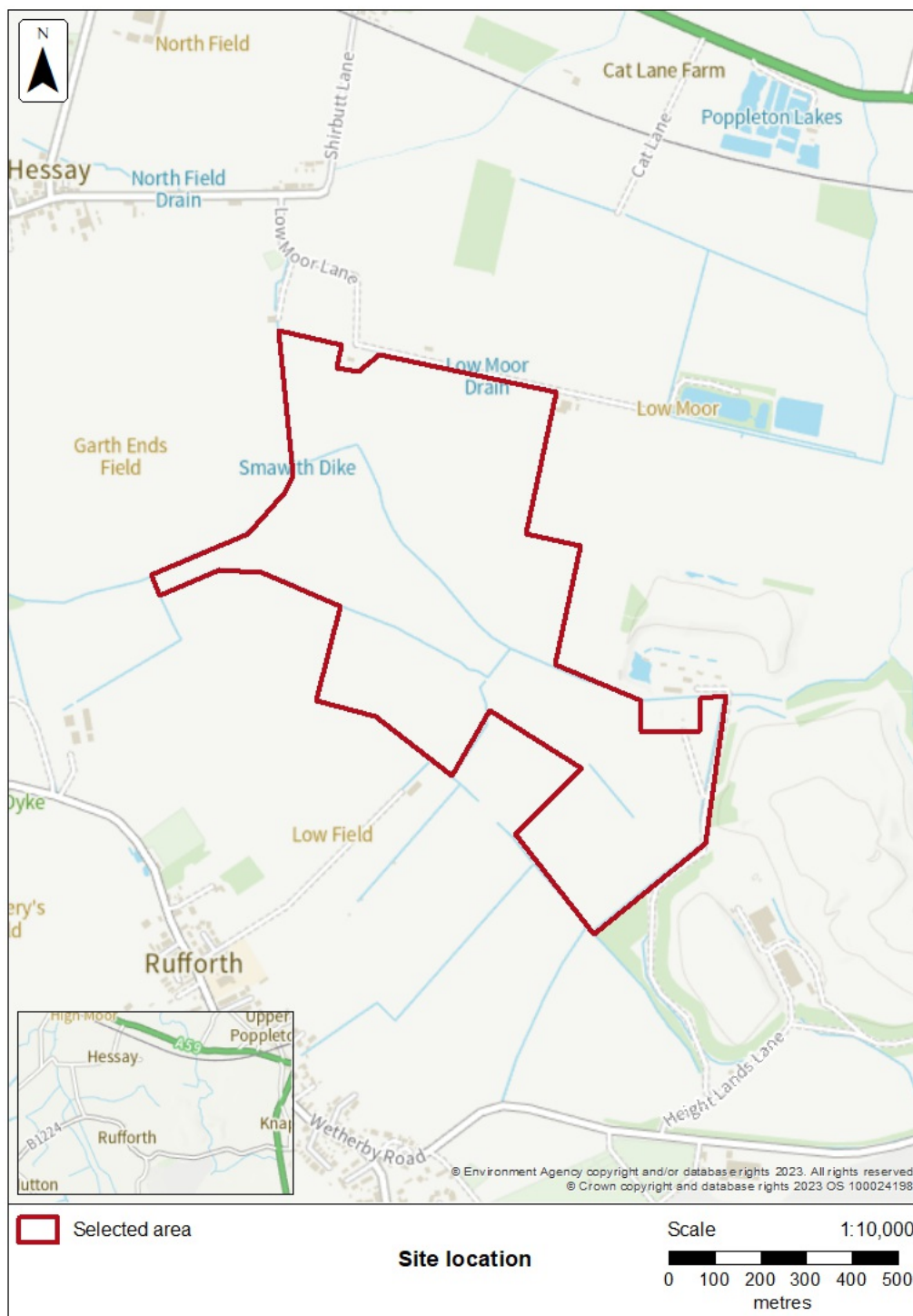
Location of site: 453244 / 452467 (shown as easting and northing coordinates)

Document created on: 19 January 2023

This information was previously known as a product 4.

Customer reference number: D6KFA4XUAN47

Map showing the location that flood risk assessment data has been requested for.



How to use this information

You can use this information as part of a flood risk assessment for a planning application. To do this, you should include it in the appendix of your flood risk assessment.

We recommend that you work with a flood risk consultant to get your flood risk assessment.

Included in this document

In this document you'll find:

- how to find information about surface water and other sources of flooding
- information on the models used
- definitions for the terminology used throughout
- flood map for planning (rivers and the sea)
- modelled data
- information about strategic flood risk assessments
- information about this data
- information about flood risk activity permits
- help and advice

Not included in this document

This document does not include a Flood Defence Breach Hazard Map.

If your location has a reduced flood risk from rivers and sea because of defences, you need to request a Flood Defence Breach Hazard Map and information about the level of flood protection offered at your location from the Yorkshire Environment Agency team at neyorkshire@environment-agency.gov.uk. This information will only be available if modelling has been carried out for breach scenarios.

Include a site location map in your request.

Information that's unavailable

This document **does not** contain:

- historic flooding
- flood defences and attributes
- climate change modelled data

We do not have historic flooding data for this location.

Please note that:

- flooding may have occurred that we do not have records for
- flooding can come from a range of different sources
- we can only supply flood risk data relating to flooding from rivers or the sea

You can contact your Lead Local Flood Authority or Internal Drainage Board to see if they

have other relevant local flood information. Please note that some areas do not have an Internal Drainage Board.

We aren't able to display flood defence locations and attributes as there are no formal flood defences in the area of interest.

There is not any modelled climate change data for this location. This is because detailed modelling hasn't been carried out in this area. You will need to consider the [latest flood risk assessment climate change allowances](#) and factor in the new allowances to demonstrate the development will be safe from flooding.

Surface water and other sources of flooding

Use the [long term flood risk service](#) to find out about the risk of flooding from:

- surface water
- ordinary watercourses
- reservoirs

For information about sewer flooding, contact the relevant water company for the area.

About the models used

Model name: Ouse CFMP Improved Jflow Outlines River Swale 9

Scenario(s): No defences exist fluvial

Date: 31 December 2008

This model contains the most relevant data for your area of interest.

Terminology used

Annual exceedance probability (AEP)

This refers to the probability of a flood event occurring in any year. The probability is expressed as a percentage. For example, a large flood which is calculated to have a 1% chance of occurring in any one year, is described as 1% AEP.

Metres above ordnance datum (mAOD)

All flood levels are given in metres above ordnance datum which is defined as the mean sea level at Newlyn, Cornwall.

Flood map for planning (rivers and the sea)

Your selected location is in flood zone 3.

Flood zone 3 shows the area at risk of flooding for an undefended flood event with a:

- 0.5% or greater probability of occurring in any year for flooding from the sea
- 1% or greater probability of occurring in any year for fluvial (river) flooding

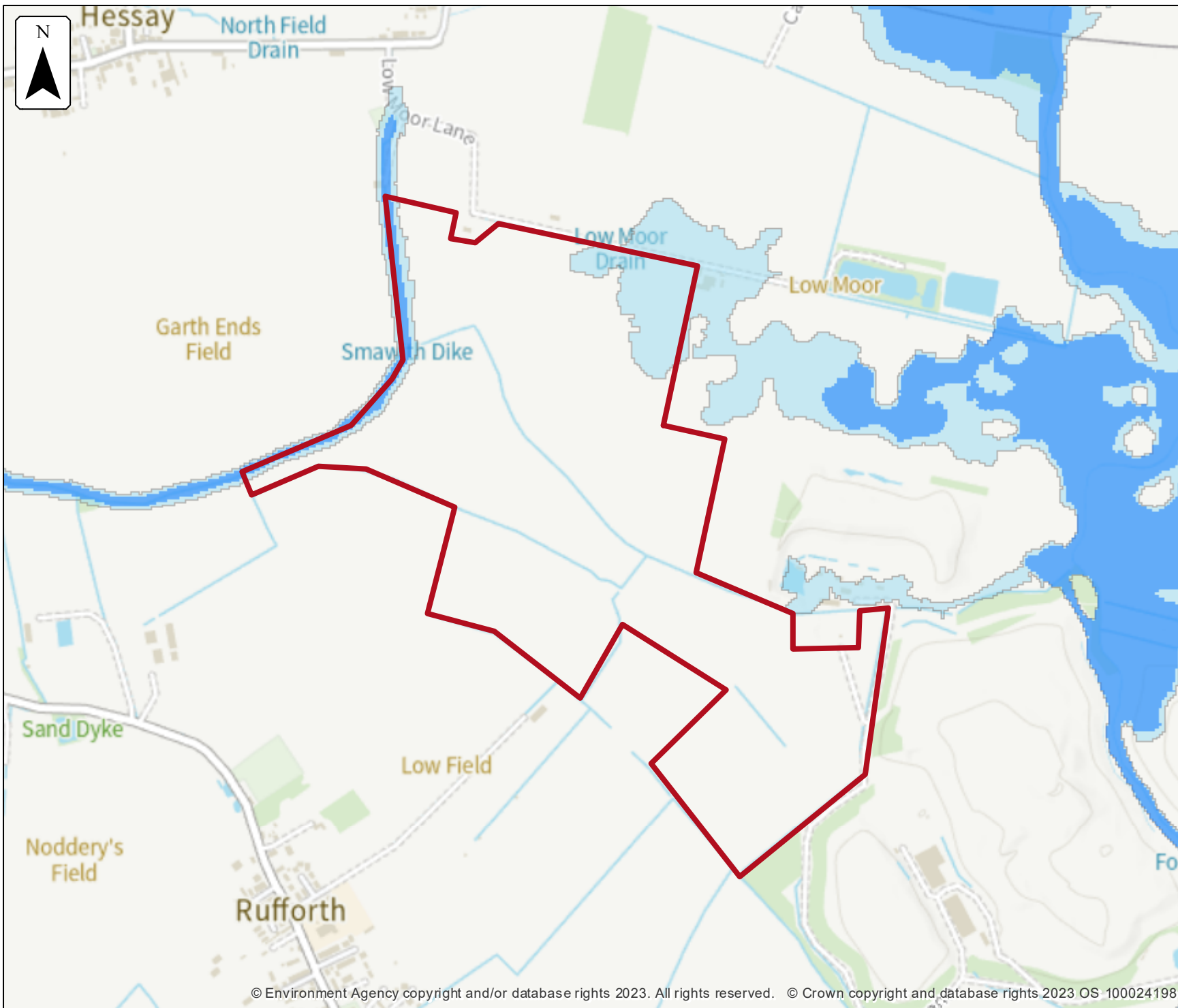
Flood zone 2 shows the area at risk of flooding for an undefended flood event with:

- between a 0.1% and 0.5% probability of occurring in any year for flooding from the sea
- between a 0.1% and 1% probability of occurring in any year for fluvial (river) flooding

It's important to remember that the flood zones on this map:

- refer to the land at risk of flooding and do not refer to individual properties
- refer to the probability of river and sea flooding, ignoring the presence of defences
- do not take into account potential impacts of climate change

This data is updated on a quarterly basis as better data becomes available.



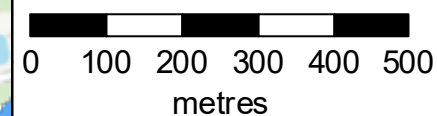
Flood map for planning

Location (easting/northing)
453244/452467

Scale
1:10,000

Created
19 Jan 2023

-  Selected area
-  Flood zone 3
-  Flood zone 2



Modelled data

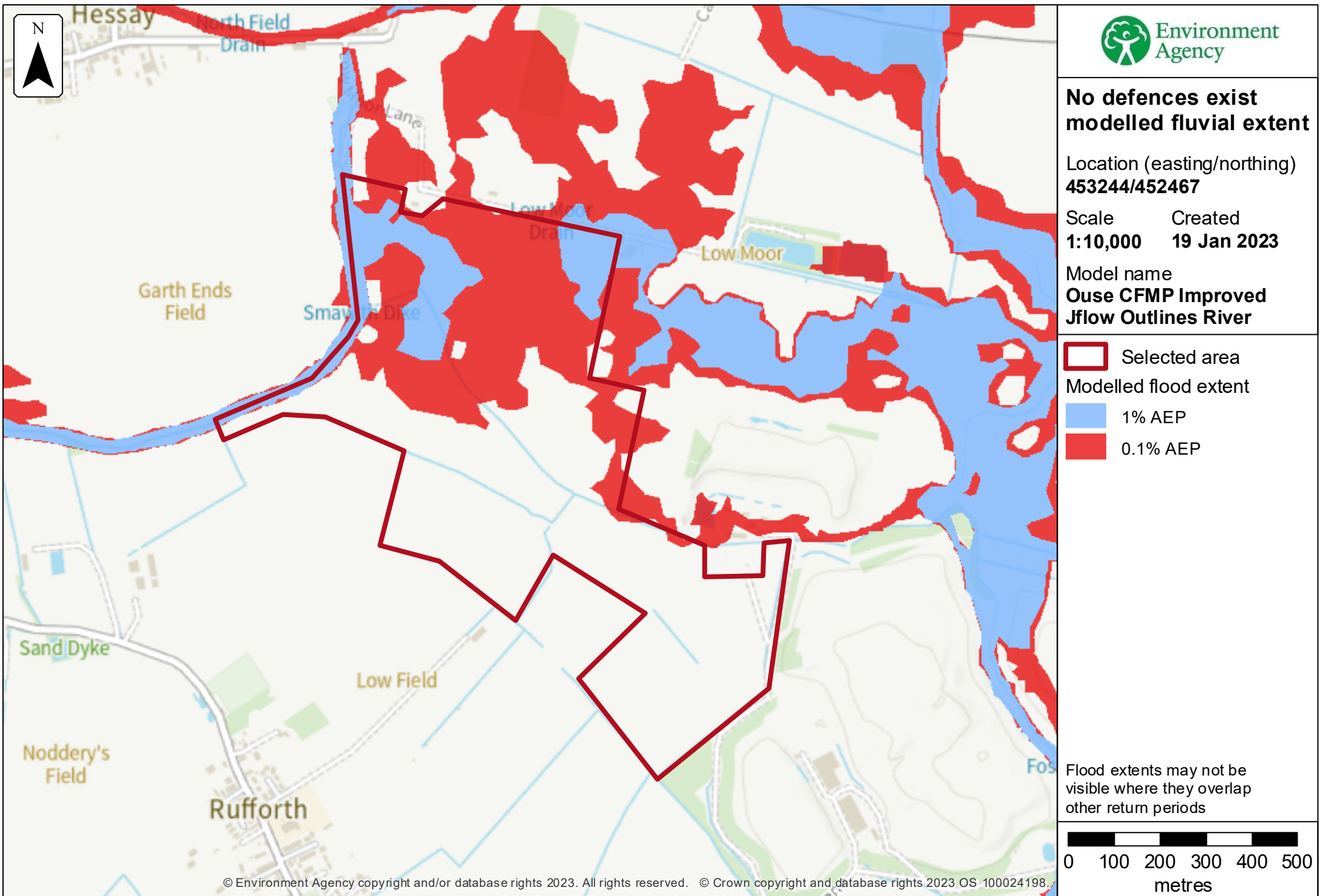
This section provides details of different scenarios we have modelled and includes the following (where available):

- outline maps showing the area at risk from flooding in different modelled scenarios

Modelled scenarios

The following scenarios are included:

- No defences exist modelled fluvial: risk of flooding from rivers where there are no flood defences



Strategic flood risk assessments

We recommend that you check the relevant local authority's strategic flood risk assessment (SFRA) as part of your work to prepare a site specific flood risk assessment.

This should give you information about:

- the potential impacts of climate change in this catchment
- areas defined as functional floodplain
- flooding from other sources, such as surface water, ground water and reservoirs

About this data

This data has been generated by strategic scale flood models and is not intended for use at the individual property scale. If you're intending to use this data as part of a flood risk assessment, please include an appropriate modelling tolerance as part of your assessment. The Environment Agency regularly updates its modelling. We recommend that you check the data provided is the most recent, before submitting your flood risk assessment.

Flood risk activity permits

Under the Environmental Permitting (England and Wales) Regulations 2016 some developments may require an environmental permit for flood risk activities from the Environment Agency. This includes any permanent or temporary works that are in, over, under, or nearby a designated main river or flood defence structure.

[Find out more about flood risk activity permits](#)

Help and advice

Contact the Yorkshire Environment Agency team at neyorkshire@environment-agency.gov.uk for:

- [more information about getting a product 5, 6, 7 or 8](#)
- general help and advice about the site you're requesting data for

APPENDIX D

Greenfield Runoff Calculations

Weetwood		Page 1
Suite 1 Park House Broncoed Bus Park Wrexham Rd Mold		
Date 14/02/2023 08:41 File	Designed by ModellingMold Checked by	
Micro Drainage		Source Control 2020.1
<u>ICP SUDS Mean Annual Flood</u> Input Return Period (years) 1000 SAAR (mm) 641 Urban 0.000 Area (ha) 1.000 Soil 0.450 Region Number Region 3 Results l/s QBAR Rural 4.0 QBAR Urban 4.0 Q1000 years 12.0 Q1 year 3.4 Q30 years 7.0 Q100 years 8.2		
©1982-2020 Innovyze		

APPENDIX E

Surface Water Attenuation - Storage Volume Calculations



Micro
Drainage

Summary of Results for 100 year Return Period (+45%)

	Storm Event	Max Level (m)	Max Depth (m)	Max Infiltration (l/s)	Max Control (l/s)	Max Σ Outflow (l/s)	Max Volume (m³)	Status
8640 min	Summer	99.598	0.048	0.0	0.9	0.9	21.1	O K
10080 min	Summer	99.581	0.031	0.0	0.9	0.9	13.5	O K
15 min	Winter	99.638	0.088	0.0	0.9	0.9	38.8	O K
30 min	Winter	99.670	0.120	0.0	0.9	0.9	53.0	O K
60 min	Winter	99.703	0.153	0.0	0.9	0.9	67.2	Flood Risk
120 min	Winter	99.734	0.184	0.0	0.9	0.9	80.9	Flood Risk
180 min	Winter	99.750	0.200	0.0	0.9	0.9	88.0	Flood Risk
240 min	Winter	99.759	0.209	0.0	0.9	0.9	92.1	Flood Risk
360 min	Winter	99.769	0.219	0.0	0.9	0.9	96.3	Flood Risk
480 min	Winter	99.773	0.223	0.0	0.9	0.9	98.2	Flood Risk
600 min	Winter	99.774	0.224	0.0	0.9	0.9	98.7	Flood Risk
720 min	Winter	99.773	0.223	0.0	0.9	0.9	98.2	Flood Risk
960 min	Winter	99.767	0.217	0.0	0.9	0.9	95.6	Flood Risk
1440 min	Winter	99.751	0.201	0.0	0.9	0.9	88.5	Flood Risk
2160 min	Winter	99.730	0.180	0.0	0.9	0.9	79.1	Flood Risk
2880 min	Winter	99.708	0.158	0.0	0.9	0.9	69.6	Flood Risk
4320 min	Winter	99.667	0.117	0.0	0.9	0.9	51.6	O K
5760 min	Winter	99.631	0.081	0.0	0.9	0.9	35.5	O K

	Storm Event	Rain (mm/hr)	Flooded Volume (m ³)	Discharge Volume (m ³)	Time-Peak (mins)
8640 min	Summer	1.125	0.0	166.5	4928
10080 min	Summer	0.991	0.0	167.8	5552
15 min	Winter	135.261	0.0	39.3	19
30 min	Winter	88.891	0.0	54.0	33
60 min	Winter	55.699	0.0	69.5	62
120 min	Winter	33.759	0.0	85.7	122
180 min	Winter	24.870	0.0	95.3	180
240 min	Winter	19.910	0.0	102.1	238
360 min	Winter	14.452	0.0	111.6	354
480 min	Winter	11.522	0.0	118.7	470
600 min	Winter	9.658	0.0	124.4	582
720 min	Winter	8.357	0.0	129.2	692
960 min	Winter	6.647	0.0	136.9	904
1440 min	Winter	4.806	0.0	137.0	1126
2160 min	Winter	3.469	0.0	158.9	1580
2880 min	Winter	2.750	0.0	166.7	2044
4320 min	Winter	1.980	0.0	176.9	2896
5760 min	Winter	1.567	0.0	183.4	3688

Weetwood		Page 4
Joseph's Well	5767	
Hanover Walk	Hessay Solar Farm	
Leeds, LS3 1AB	Drainage Area 1a	
Date 23/02/2023 16:58	Designed by TB	
File 2023-02-22 5767 Hydro-Br...	Checked by	
XP Solutions		Source Control 2020.1

Rainfall Details

Rainfall Model	FSR	Winter Storms	Yes
Return Period (years)	100	Cv (Summer)	0.750
Region	England and Wales	Cv (Winter)	0.840
M5-60 (mm)	19.000	Shortest Storm (mins)	15
Ratio R	0.400	Longest Storm (mins)	10080
Summer Storms	Yes	Climate Change %	+45

Time Area Diagram

Total Area (ha) 0.165

Time (mins)	Area
From:	To: (ha)
0	4 0.165

©1982-2020 Innovyze

Weetwood		Page 5
Joseph's Well	5767	
Hanover Walk	Hessay Solar Farm	
Leeds, LS3 1AB	Drainage Area 1a	
Date 23/02/2023 16:58	Designed by TB	
File 2023-02-22 5767 Hydro-Br...	Checked by	
XP Solutions		Source Control 2020.1

Model Details

Storage is Online Cover Level (m) 100.000

Porous Car Park Structure

Infiltration Coefficient Base (m/hr)	0.00000	Width (m)	4.5
Membrane Percolation (mm/hr)	1000	Length (m)	326.0
Max Percolation (l/s)	407.5	Slope (1:X)	0.0
Safety Factor	2.0	Depression Storage (mm)	5
Porosity	0.30	Evaporation (mm/day)	3
Invert Level (m)	99.550	Cap Volume Depth (m)	0.300

Hydro-Brake® Optimum Outflow Control

Unit Reference	MD-SHE-0050-1000-0750-1000
Design Head (m)	0.750
Design Flow (l/s)	1.0
Flush-Flo™	Calculated
Objective	Minimise upstream storage
Application	Surface
Sump Available	Yes
Diameter (mm)	50
Invert Level (m)	99.150
Minimum Outlet Pipe Diameter (mm)	75
Suggested Manhole Diameter (mm)	1200

Control Points	Head (m)	Flow (l/s)	Control Points	Head (m)	Flow (l/s)
Design Point (Calculated)	0.750	1.0	Kick-Flo®	0.445	0.8
Flush-Flo™	0.222	1.0	Mean Flow over Head Range	-	0.9

The hydrological calculations have been based on the Head/Discharge relationship for the Hydro-Brake® Optimum as specified. Should another type of control device other than a Hydro-Brake Optimum® be utilised then these storage routing calculations will be invalidated

Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)
0.100	0.9	1.200	1.2	3.000	1.9	7.000	2.8
0.200	1.0	1.400	1.3	3.500	2.0	7.500	2.9
0.300	0.9	1.600	1.4	4.000	2.1	8.000	2.9
0.400	0.9	1.800	1.5	4.500	2.3	8.500	3.0
0.500	0.8	2.000	1.6	5.000	2.4	9.000	3.1
0.600	0.9	2.200	1.6	5.500	2.5	9.500	3.2
0.800	1.0	2.400	1.7	6.000	2.6		
1.000	1.1	2.600	1.7	6.500	2.7		

©1982-2020 Innovyze



Micro
Drainage

Half Drain Time : 2059 minutes.

Storm Event	Rain (mm/hr)	Flooded Volume (m ³)	Discharge Volume (m ³)	Time-Peak (mins)
----------------	-----------------	--	--	---------------------

©1982-2020 Innovyze

Weetwood		Page 4
Joseph's Well	5767	
Hanover Walk	Hessay Solar Farm	
Leeds, LS3 1AB	Drainage Area 1b	
Date 23/02/2023 16:59	Designed by TB	
File 2023-02-22 5767 Hydro-Br...	Checked by	
XP Solutions		Source Control 2020.1

Rainfall Details

Rainfall Model	FSR	Winter Storms	Yes
Return Period (years)	100	Cv (Summer)	0.750
Region	England and Wales	Cv (Winter)	0.840
M5-60 (mm)	19.000	Shortest Storm (mins)	15
Ratio R	0.400	Longest Storm (mins)	10080
Summer Storms	Yes	Climate Change %	+45

Time Area Diagram

Total Area (ha) 0.319

Time (mins)	Area
From:	To: (ha)
0	4 0.319

©1982-2020 Innovyze

Weetwood		Page 5
Joseph's Well	5767	
Hanover Walk	Hessay Solar Farm	
Leeds, LS3 1AB	Drainage Area 1b	
Date 23/02/2023 16:59	Designed by TB	
File 2023-02-22 5767 Hydro-Br...	Checked by	
XP Solutions		Source Control 2020.1

Model Details

Storage is Online Cover Level (m) 100.000

Porous Car Park Structure

Infiltration Coefficient Base (m/hr)	0.00000	Width (m)	4.5
Membrane Percolation (mm/hr)	1000	Length (m)	610.0
Max Percolation (l/s)	762.5	Slope (1:X)	0.0
Safety Factor	2.0	Depression Storage (mm)	5
Porosity	0.30	Evaporation (mm/day)	3
Invert Level (m)	99.550	Cap Volume Depth (m)	0.300

Hydro-Brake® Optimum Outflow Control

Unit Reference	MD-SHE-0050-1000-0750-1000
Design Head (m)	0.750
Design Flow (l/s)	1.0
Flush-Flo™	Calculated
Objective	Minimise upstream storage
Application	Surface
Sump Available	Yes
Diameter (mm)	50
Invert Level (m)	99.150
Minimum Outlet Pipe Diameter (mm)	75
Suggested Manhole Diameter (mm)	1200

Control Points	Head (m)	Flow (l/s)	Control Points	Head (m)	Flow (l/s)
Design Point (Calculated)	0.750	1.0	Kick-Flo®	0.445	0.8
Flush-Flo™	0.222	1.0	Mean Flow over Head Range	-	0.9

The hydrological calculations have been based on the Head/Discharge relationship for the Hydro-Brake® Optimum as specified. Should another type of control device other than a Hydro-Brake Optimum® be utilised then these storage routing calculations will be invalidated

Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)
0.100	0.9	1.200	1.2	3.000	1.9	7.000	2.8
0.200	1.0	1.400	1.3	3.500	2.0	7.500	2.9
0.300	0.9	1.600	1.4	4.000	2.1	8.000	2.9
0.400	0.9	1.800	1.5	4.500	2.3	8.500	3.0
0.500	0.8	2.000	1.6	5.000	2.4	9.000	3.1
0.600	0.9	2.200	1.6	5.500	2.5	9.500	3.2
0.800	1.0	2.400	1.7	6.000	2.6		
1.000	1.1	2.600	1.7	6.500	2.7		

©1982-2020 Innovyze



**Micro
Drainage**

Half Drain Time : 2435 minutes.

Weetwood		Page 5
Joseph's Well	5767	
Hanover Walk	Hessay Solar Farm	
Leeds, LS3 1AB	Drainage Area 2	
Date 23/02/2023 17:02	Designed by TB	
File 2023-02-22 5767 Hydro-Br...	Checked by	
XP Solutions	Source Control 2020.1	

Model Details

Storage is Online Cover Level (m) 100.000

Porous Car Park Structure

Infiltration Coefficient Base (m/hr)	0.00000	Width (m)	4.5
Membrane Percolation (mm/hr)	1000	Length (m)	507.0
Max Percolation (l/s)	633.8	Slope (1:X)	0.0
Safety Factor	2.0	Depression Storage (mm)	5
Porosity	0.30	Evaporation (mm/day)	3
Invert Level (m)	99.450	Cap Volume Depth (m)	0.400

Hydro-Brake® Optimum Outflow Control

Unit Reference	MD-SHE-0050-1000-0750-1000
Design Head (m)	0.750
Design Flow (l/s)	1.0
Flush-Flo™	Calculated
Objective	Minimise upstream storage
Application	Surface
Sump Available	Yes
Diameter (mm)	50
Invert Level (m)	99.150
Minimum Outlet Pipe Diameter (mm)	75
Suggested Manhole Diameter (mm)	1200

Control Points	Head (m)	Flow (l/s)	Control Points	Head (m)	Flow (l/s)
Design Point (Calculated)	0.750	1.0	Kick-Flo®	0.445	0.8
Flush-Flo™	0.222	1.0	Mean Flow over Head Range	-	0.9

The hydrological calculations have been based on the Head/Discharge relationship for the Hydro-Brake® Optimum as specified. Should another type of control device other than a Hydro-Brake Optimum® be utilised then these storage routing calculations will be invalidated

Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)
0.100	0.9	1.200	1.2	3.000	1.9	7.000	2.8
0.200	1.0	1.400	1.3	3.500	2.0	7.500	2.9
0.300	0.9	1.600	1.4	4.000	2.1	8.000	2.9
0.400	0.9	1.800	1.5	4.500	2.3	8.500	3.0
0.500	0.8	2.000	1.6	5.000	2.4	9.000	3.1
0.600	0.9	2.200	1.6	5.500	2.5	9.500	3.2
0.800	1.0	2.400	1.7	6.000	2.6		
1.000	1.1	2.600	1.7	6.500	2.7		

©1982-2020 Innovyze



Half Drain Time : 830 minutes.

Storm Event			Max Level	Max Depth	Max Infiltration	Max Control	Max Outflow	Max Volume	Status
			(m)	(m)	(l/s)	(l/s)	(l/s)	(m³)	
15 min	Summer		99.632	0.082	0.0	0.9	0.9	30.6	O K
30 min	Summer		99.661	0.111	0.0	0.9	0.9	41.8	O K
60 min	Summer		99.691	0.141	0.0	0.9	0.9	53.0	O K
120 min	Summer		99.719	0.169	0.0	0.9	0.9	63.3	Flood Risk
180 min	Summer		99.733	0.183	0.0	0.9	0.9	68.5	Flood Risk
240 min	Summer		99.740	0.190	0.0	0.9	0.9	71.3	Flood Risk
360 min	Summer		99.746	0.196	0.0	0.9	0.9	73.6	Flood Risk
480 min	Summer		99.747	0.197	0.0	0.9	0.9	74.0	Flood Risk
600 min	Summer		99.745	0.195	0.0	0.9	0.9	73.2	Flood Risk
720 min	Summer		99.741	0.191	0.0	0.9	0.9	71.8	Flood Risk
960 min	Summer		99.734	0.184	0.0	0.9	0.9	68.9	Flood Risk
1440 min	Summer		99.720	0.170	0.0	0.9	0.9	63.9	Flood Risk
2160 min	Summer		99.703	0.153	0.0	0.9	0.9	57.4	Flood Risk
2880 min	Summer		99.686	0.136	0.0	0.9	0.9	51.2	O K
4320 min	Summer		99.656	0.106	0.0	0.9	0.9	39.8	O K
5760 min	Summer		99.629	0.079	0.0	0.9	0.9	29.7	O K
7200 min	Summer		99.606	0.056	0.0	0.9	0.9	21.0	O K

Storm			Rain	Flooded	Discharge	Time-Peak
Event			(mm/hr)	Volume	Volume	(mins)
				(m ³)	(m ³)	
15 min	Summer		135.261	0.0	31.1	19
30 min	Summer		88.891	0.0	42.8	34
60 min	Summer		55.699	0.0	55.3	64
120 min	Summer		33.759	0.0	68.2	122
180 min	Summer		24.870	0.0	75.9	182
240 min	Summer		19.910	0.0	81.3	242
360 min	Summer		14.452	0.0	88.9	362
480 min	Summer		11.522	0.0	94.6	480
600 min	Summer		9.658	0.0	99.1	600
720 min	Summer		8.357	0.0	103.0	656
960 min	Summer		6.647	0.0	109.1	768
1440 min	Summer		4.806	0.0	117.9	1024
2160 min	Summer		3.469	0.0	126.5	1428
2880 min	Summer		2.750	0.0	132.5	1844
4320 min	Summer		1.980	0.0	140.5	2636
5760 min	Summer		1.567	0.0	145.4	3400
7200 min	Summer		1.306	0.0	148.7	4112



Micro
Drainage

Summary of Results for 100 year Return Period (+45%)

	Storm Event	Max Level (m)	Max Depth (m)	Max Infiltration (l/s)	Max Control (l/s)	Max Σ Outflow (l/s)	Max Volume (m³)	Status
8640 min	Summer	99.585	0.035	0.0	0.9	0.9	13.2	O K
10080 min	Summer	99.569	0.019	0.0	0.9	0.9	7.2	O K
15 min	Winter	99.643	0.093	0.0	0.9	0.9	35.1	O K
30 min	Winter	99.677	0.127	0.0	0.9	0.9	47.7	O K
60 min	Winter	99.711	0.161	0.0	0.9	0.9	60.4	Flood Risk
120 min	Winter	99.743	0.193	0.0	0.9	0.9	72.3	Flood Risk
180 min	Winter	99.759	0.209	0.0	0.9	0.9	78.4	Flood Risk
240 min	Winter	99.768	0.218	0.0	0.9	0.9	81.8	Flood Risk
360 min	Winter	99.776	0.226	0.0	0.9	0.9	85.0	Flood Risk
480 min	Winter	99.780	0.230	0.0	0.9	0.9	86.2	Flood Risk
600 min	Winter	99.779	0.229	0.0	0.9	0.9	86.0	Flood Risk
720 min	Winter	99.777	0.227	0.0	0.9	0.9	85.1	Flood Risk
960 min	Winter	99.768	0.218	0.0	0.9	0.9	81.7	Flood Risk
1440 min	Winter	99.750	0.200	0.0	0.9	0.9	75.2	Flood Risk
2160 min	Winter	99.725	0.175	0.0	0.9	0.9	65.8	Flood Risk
2880 min	Winter	99.701	0.151	0.0	0.9	0.9	56.6	Flood Risk
4320 min	Winter	99.655	0.105	0.0	0.9	0.9	39.4	O K
5760 min	Winter	99.615	0.065	0.0	0.9	0.9	24.4	O K

	Storm Event	Rain (mm/hr)	Flooded Volume (m ³)	Discharge Volume (m ³)	Time-Peak (mins)
8640 min	Summer	1.125	0.0	150.7	4832
10080 min	Summer	0.991	0.0	152.2	5448
15 min	Winter	135.261	0.0	35.6	19
30 min	Winter	88.891	0.0	48.7	33
60 min	Winter	55.699	0.0	62.6	62
120 min	Winter	33.759	0.0	77.2	122
180 min	Winter	24.870	0.0	85.8	180
240 min	Winter	19.910	0.0	92.0	238
360 min	Winter	14.452	0.0	100.4	354
480 min	Winter	11.522	0.0	106.8	468
600 min	Winter	9.658	0.0	112.0	578
720 min	Winter	8.357	0.0	116.3	686
960 min	Winter	6.647	0.0	123.2	886
1440 min	Winter	4.806	0.0	133.2	1098
2160 min	Winter	3.469	0.0	143.2	1556
2880 min	Winter	2.750	0.0	150.2	1992
4320 min	Winter	1.980	0.0	159.6	2848
5760 min	Winter	1.567	0.0	165.7	3624

Weetwood		Page 5
Joseph's Well	5767	
Hanover Walk	Hessay Solar Farm	
Leeds, LS3 1AB	Drainage Area 3	
Date 23/02/2023 17:03	Designed by TB	
File 2023-02-22 5767 Hydro-Br...	Checked by	
XP Solutions	Source Control 2020.1	

Model Details

Storage is Online Cover Level (m) 100.000

Porous Car Park Structure

Infiltration Coefficient Base (m/hr)	0.00000	Width (m)	4.5
Membrane Percolation (mm/hr)	1000	Length (m)	278.0
Max Percolation (l/s)	347.5	Slope (1:X)	0.0
Safety Factor	2.0	Depression Storage (mm)	5
Porosity	0.30	Evaporation (mm/day)	3
Invert Level (m)	99.550	Cap Volume Depth (m)	0.300

Hydro-Brake® Optimum Outflow Control

Unit Reference	MD-SHE-0050-1000-0750-1000
Design Head (m)	0.750
Design Flow (l/s)	1.0
Flush-Flo™	Calculated
Objective	Minimise upstream storage
Application	Surface
Sump Available	Yes
Diameter (mm)	50
Invert Level (m)	99.150
Minimum Outlet Pipe Diameter (mm)	75
Suggested Manhole Diameter (mm)	1200

Control Points	Head (m)	Flow (l/s)	Control Points	Head (m)	Flow (l/s)
Design Point (Calculated)	0.750	1.0	Kick-Flo®	0.445	0.8
Flush-Flo™	0.222	1.0	Mean Flow over Head Range	-	0.9

The hydrological calculations have been based on the Head/Discharge relationship for the Hydro-Brake® Optimum as specified. Should another type of control device other than a Hydro-Brake Optimum® be utilised then these storage routing calculations will be invalidated

Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)
0.100	0.9	1.200	1.2	3.000	1.9	7.000	2.8
0.200	1.0	1.400	1.3	3.500	2.0	7.500	2.9
0.300	0.9	1.600	1.4	4.000	2.1	8.000	2.9
0.400	0.9	1.800	1.5	4.500	2.3	8.500	3.0
0.500	0.8	2.000	1.6	5.000	2.4	9.000	3.1
0.600	0.9	2.200	1.6	5.500	2.5	9.500	3.2
0.800	1.0	2.400	1.7	6.000	2.6		
1.000	1.1	2.600	1.7	6.500	2.7		

©1982-2020 Innovyze

Weetwood		Page 5
Joseph's Well	5767	
Hanover Walk	Hessay Solar Farm	
Leeds, LS3 1AB	Drainage Area 4a	
Date 23/02/2023 17:03	Designed by TB	
File 2023-02-22 5767 Hydro-Br...	Checked by	
XP Solutions		Source Control 2020.1

Model Details

Storage is Online Cover Level (m) 100.000

Porous Car Park Structure

Infiltration Coefficient Base (m/hr)	0.00000	Width (m)	4.5
Membrane Percolation (mm/hr)	1000	Length (m)	314.0
Max Percolation (l/s)	392.5	Slope (1:X)	0.0
Safety Factor	2.0	Depression Storage (mm)	5
Porosity	0.30	Evaporation (mm/day)	3
Invert Level (m)	99.550	Cap Volume Depth (m)	0.300

Hydro-Brake® Optimum Outflow Control

Unit Reference	MD-SHE-0050-1000-0750-1000
Design Head (m)	0.750
Design Flow (l/s)	1.0
Flush-Flo™	Calculated
Objective	Minimise upstream storage
Application	Surface
Sump Available	Yes
Diameter (mm)	50
Invert Level (m)	99.150
Minimum Outlet Pipe Diameter (mm)	75
Suggested Manhole Diameter (mm)	1200

Control Points	Head (m)	Flow (l/s)	Control Points	Head (m)	Flow (l/s)
Design Point (Calculated)	0.750	1.0	Kick-Flo®	0.445	0.8
Flush-Flo™	0.222	1.0	Mean Flow over Head Range	-	0.9

The hydrological calculations have been based on the Head/Discharge relationship for the Hydro-Brake® Optimum as specified. Should another type of control device other than a Hydro-Brake Optimum® be utilised then these storage routing calculations will be invalidated

Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)
0.100	0.9	1.200	1.2	3.000	1.9	7.000	2.8
0.200	1.0	1.400	1.3	3.500	2.0	7.500	2.9
0.300	0.9	1.600	1.4	4.000	2.1	8.000	2.9
0.400	0.9	1.800	1.5	4.500	2.3	8.500	3.0
0.500	0.8	2.000	1.6	5.000	2.4	9.000	3.1
0.600	0.9	2.200	1.6	5.500	2.5	9.500	3.2
0.800	1.0	2.400	1.7	6.000	2.6		
1.000	1.1	2.600	1.7	6.500	2.7		

©1982-2020 Innovyze



Summary of Results for 100 year Return Period (+45%)

	Storm Event	Max Level (m)	Max Depth (m)	Max Infiltration (l/s)	Max Control (l/s)	Max Σ Outflow (l/s)	Max Volume (m³)	Status
8640 min	Summer	99.651	0.101	0.0	0.9	0.9	59.4	O K
10080 min	Summer	99.634	0.084	0.0	0.9	0.9	49.5	O K
15 min	Winter	99.646	0.096	0.0	0.9	0.9	56.0	O K
30 min	Winter	99.680	0.130	0.0	0.9	0.9	76.2	O K
60 min	Winter	99.715	0.165	0.0	0.9	0.9	97.0	Flood Risk
120 min	Winter	99.750	0.200	0.0	0.9	0.9	117.4	Flood Risk
180 min	Winter	99.769	0.219	0.0	0.9	0.9	128.5	Flood Risk
240 min	Winter	99.781	0.231	0.0	0.9	0.9	135.4	Flood Risk
360 min	Winter	99.795	0.245	0.0	0.9	0.9	143.4	Flood Risk
480 min	Winter	99.803	0.253	0.0	0.9	0.9	148.2	Flood Risk
600 min	Winter	99.808	0.258	0.0	0.9	0.9	150.9	Flood Risk
720 min	Winter	99.810	0.260	0.0	0.9	0.9	152.3	Flood Risk
960 min	Winter	99.810	0.260	0.0	0.9	0.9	152.4	Flood Risk
1440 min	Winter	99.801	0.251	0.0	0.9	0.9	147.0	Flood Risk
2160 min	Winter	99.781	0.231	0.0	0.9	0.9	135.5	Flood Risk
2880 min	Winter	99.765	0.215	0.0	0.9	0.9	125.8	Flood Risk
4320 min	Winter	99.731	0.181	0.0	0.9	0.9	106.1	Flood Risk
5760 min	Winter	99.699	0.149	0.0	0.9	0.9	87.2	O K

	Storm Event	Rain (mm/hr)	Flooded Volume (m ³)	Discharge Volume (m ³)	Time-Peak (mins)
8640 min	Summer	1.125	0.0	239.2	5184
10080 min	Summer	0.991	0.0	241.1	5944
15 min	Winter	135.261	0.0	56.4	19
30 min	Winter	88.891	0.0	71.0	33
60 min	Winter	55.699	0.0	99.2	64
120 min	Winter	33.759	0.0	122.1	122
180 min	Winter	24.870	0.0	135.8	180
240 min	Winter	19.910	0.0	145.6	240
360 min	Winter	14.452	0.0	145.9	356
480 min	Winter	11.522	0.0	146.1	472
600 min	Winter	9.658	0.0	146.3	588
720 min	Winter	8.357	0.0	146.3	702
960 min	Winter	6.647	0.0	145.7	926
1440 min	Winter	4.806	0.0	143.3	1354
2160 min	Winter	3.469	0.0	226.7	1688
2880 min	Winter	2.750	0.0	237.8	2136
4320 min	Winter	1.980	0.0	252.7	3028
5760 min	Winter	1.567	0.0	262.3	3920

Weetwood		Page 5																																																																																																																																								
Joseph's Well	5767																																																																																																																																									
Hanover Walk	Hessay Solar Farm																																																																																																																																									
Leeds, LS3 1AB	Drainage Area 4b																																																																																																																																									
Date 23/02/2023 17:05	Designed by TB																																																																																																																																									
File 2023-02-22 5767 Hydro-Br...	Checked by																																																																																																																																									
XP Solutions	Source Control 2020.1																																																																																																																																									
<u>Model Details</u> Storage is Online Cover Level (m) 100.000 <u>Porous Car Park Structure</u> <table><tr><td>Infiltration Coefficient Base (m/hr)</td><td>0.00000</td><td>Width (m)</td><td>4.5</td></tr><tr><td>Membrane Percolation (mm/hr)</td><td>1000</td><td>Length (m)</td><td>434.0</td></tr><tr><td>Max Percolation (l/s)</td><td>542.5</td><td>Slope (1:X)</td><td>0.0</td></tr><tr><td>Safety Factor</td><td>2.0</td><td>Depression Storage (mm)</td><td>5</td></tr><tr><td>Porosity</td><td>0.30</td><td>Evaporation (mm/day)</td><td>3</td></tr><tr><td>Invert Level (m)</td><td>99.550</td><td>Cap Volume Depth (m)</td><td>0.300</td></tr></table> <u>Hydro-Brake® Optimum Outflow Control</u> <table><tr><td>Unit Reference</td><td>MD-SHE-0050-1000-0750-1000</td></tr><tr><td>Design Head (m)</td><td>0.750</td></tr><tr><td>Design Flow (l/s)</td><td>1.0</td></tr><tr><td>Flush-Flo™</td><td>Calculated</td></tr><tr><td>Objective</td><td>Minimise upstream storage</td></tr><tr><td>Application</td><td>Surface</td></tr><tr><td>Sump Available</td><td>Yes</td></tr><tr><td>Diameter (mm)</td><td>50</td></tr><tr><td>Invert Level (m)</td><td>99.150</td></tr><tr><td>Minimum Outlet Pipe Diameter (mm)</td><td>75</td></tr><tr><td>Suggested Manhole Diameter (mm)</td><td>1200</td></tr></table> <table><tr><th>Control Points</th><th>Head (m)</th><th>Flow (l/s)</th><th>Control Points</th><th>Head (m)</th><th>Flow (l/s)</th></tr><tr><td>Design Point (Calculated)</td><td>0.750</td><td>1.0</td><td>Kick-Flo®</td><td>0.445</td><td>0.8</td></tr><tr><td>Flush-Flo™</td><td>0.222</td><td>1.0</td><td>Mean Flow over Head Range</td><td>-</td><td>0.9</td></tr></table> The hydrological calculations have been based on the Head/Discharge relationship for the Hydro-Brake® Optimum as specified. Should another type of control device other than a Hydro-Brake Optimum® be utilised then these storage routing calculations will be invalidated <table><tr><th>Depth (m)</th><th>Flow (l/s)</th><th>Depth (m)</th><th>Flow (l/s)</th><th>Depth (m)</th><th>Flow (l/s)</th><th>Depth (m)</th><th>Flow (l/s)</th></tr><tr><td>0.100</td><td>0.9</td><td>1.200</td><td>1.2</td><td>3.000</td><td>1.9</td><td>7.000</td><td>2.8</td></tr><tr><td>0.200</td><td>1.0</td><td>1.400</td><td>1.3</td><td>3.500</td><td>2.0</td><td>7.500</td><td>2.9</td></tr><tr><td>0.300</td><td>0.9</td><td>1.600</td><td>1.4</td><td>4.000</td><td>2.1</td><td>8.000</td><td>2.9</td></tr><tr><td>0.400</td><td>0.9</td><td>1.800</td><td>1.5</td><td>4.500</td><td>2.3</td><td>8.500</td><td>3.0</td></tr><tr><td>0.500</td><td>0.8</td><td>2.000</td><td>1.6</td><td>5.000</td><td>2.4</td><td>9.000</td><td>3.1</td></tr><tr><td>0.600</td><td>0.9</td><td>2.200</td><td>1.6</td><td>5.500</td><td>2.5</td><td>9.500</td><td>3.2</td></tr><tr><td>0.800</td><td>1.0</td><td>2.400</td><td>1.7</td><td>6.000</td><td>2.6</td><td></td><td></td></tr><tr><td>1.000</td><td>1.1</td><td>2.600</td><td>1.7</td><td>6.500</td><td>2.7</td><td></td><td></td></tr></table>			Infiltration Coefficient Base (m/hr)	0.00000	Width (m)	4.5	Membrane Percolation (mm/hr)	1000	Length (m)	434.0	Max Percolation (l/s)	542.5	Slope (1:X)	0.0	Safety Factor	2.0	Depression Storage (mm)	5	Porosity	0.30	Evaporation (mm/day)	3	Invert Level (m)	99.550	Cap Volume Depth (m)	0.300	Unit Reference	MD-SHE-0050-1000-0750-1000	Design Head (m)	0.750	Design Flow (l/s)	1.0	Flush-Flo™	Calculated	Objective	Minimise upstream storage	Application	Surface	Sump Available	Yes	Diameter (mm)	50	Invert Level (m)	99.150	Minimum Outlet Pipe Diameter (mm)	75	Suggested Manhole Diameter (mm)	1200	Control Points	Head (m)	Flow (l/s)	Control Points	Head (m)	Flow (l/s)	Design Point (Calculated)	0.750	1.0	Kick-Flo®	0.445	0.8	Flush-Flo™	0.222	1.0	Mean Flow over Head Range	-	0.9	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	0.100	0.9	1.200	1.2	3.000	1.9	7.000	2.8	0.200	1.0	1.400	1.3	3.500	2.0	7.500	2.9	0.300	0.9	1.600	1.4	4.000	2.1	8.000	2.9	0.400	0.9	1.800	1.5	4.500	2.3	8.500	3.0	0.500	0.8	2.000	1.6	5.000	2.4	9.000	3.1	0.600	0.9	2.200	1.6	5.500	2.5	9.500	3.2	0.800	1.0	2.400	1.7	6.000	2.6			1.000	1.1	2.600	1.7	6.500	2.7		
Infiltration Coefficient Base (m/hr)	0.00000	Width (m)	4.5																																																																																																																																							
Membrane Percolation (mm/hr)	1000	Length (m)	434.0																																																																																																																																							
Max Percolation (l/s)	542.5	Slope (1:X)	0.0																																																																																																																																							
Safety Factor	2.0	Depression Storage (mm)	5																																																																																																																																							
Porosity	0.30	Evaporation (mm/day)	3																																																																																																																																							
Invert Level (m)	99.550	Cap Volume Depth (m)	0.300																																																																																																																																							
Unit Reference	MD-SHE-0050-1000-0750-1000																																																																																																																																									
Design Head (m)	0.750																																																																																																																																									
Design Flow (l/s)	1.0																																																																																																																																									
Flush-Flo™	Calculated																																																																																																																																									
Objective	Minimise upstream storage																																																																																																																																									
Application	Surface																																																																																																																																									
Sump Available	Yes																																																																																																																																									
Diameter (mm)	50																																																																																																																																									
Invert Level (m)	99.150																																																																																																																																									
Minimum Outlet Pipe Diameter (mm)	75																																																																																																																																									
Suggested Manhole Diameter (mm)	1200																																																																																																																																									
Control Points	Head (m)	Flow (l/s)	Control Points	Head (m)	Flow (l/s)																																																																																																																																					
Design Point (Calculated)	0.750	1.0	Kick-Flo®	0.445	0.8																																																																																																																																					
Flush-Flo™	0.222	1.0	Mean Flow over Head Range	-	0.9																																																																																																																																					
Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)	Depth (m)	Flow (l/s)																																																																																																																																			
0.100	0.9	1.200	1.2	3.000	1.9	7.000	2.8																																																																																																																																			
0.200	1.0	1.400	1.3	3.500	2.0	7.500	2.9																																																																																																																																			
0.300	0.9	1.600	1.4	4.000	2.1	8.000	2.9																																																																																																																																			
0.400	0.9	1.800	1.5	4.500	2.3	8.500	3.0																																																																																																																																			
0.500	0.8	2.000	1.6	5.000	2.4	9.000	3.1																																																																																																																																			
0.600	0.9	2.200	1.6	5.500	2.5	9.500	3.2																																																																																																																																			
0.800	1.0	2.400	1.7	6.000	2.6																																																																																																																																					
1.000	1.1	2.600	1.7	6.500	2.7																																																																																																																																					
©1982-2020 Innovyze																																																																																																																																										

Delivering client focussed services nationally

Flood Risk Assessments
Flood Consequences Assessments
Surface Water Drainage
Foul Water Drainage
Environmental Impact Assessments
River Realignment and Restoration
Water Framework Directive Assessments
Environmental Permit and Land Drainage Applications
Sequential, Justification and Exception Tests
Utility Assessments
Expert Witness and Planning Appeals
Discharge of Planning Conditions

www.weetwood.net